



THE UNIVERSITY OF CHENAB
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SOFTWARE REQUIREMENTS
SPECIFICATION
(SRS DOCUMENT)

for

3DPanoSphere(Your personalized 3D image generation app)
Version 1.0

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1 Introduction

3DPanoSphere is a mobile application designed to provide users with an interactive platform for creating, sharing, and exploring 360° panoramic images and videos. The app enables users to upload panoramic content, add interactive hotspots, create virtual tours, and navigate through panoramic stories. Social engagement is integrated through likes, comments, following, real-time messaging, and a star rating system to evaluate content quality. The platform focuses on immersive experiences for tourism, education, real estate, and other domains where 360° visuals enhance user understanding and interaction.

1.1 Purpose

The purpose of 3DPanoSphere is to provide an easy-to-use, engaging, and interactive platform for both content creators and viewers:

- **For creators:** Upload and manage panoramic images and videos, create virtual tours, and add interactive elements.
- **For viewers:** Explore immersive panoramic content, interact via likes, comments, and ratings, and navigate virtual tours.
- **For the platform:** Facilitate content discovery and engagement through trending sections, search functionality, and a star rating system to highlight high-quality panoramas.

The system aims to combine visual immersion with social interaction, making panoramic content accessible and engaging to a wide audience.

1.2 Scope

The scope of 3DPanoSphere is to provide an interactive platform for uploading, managing, and exploring 360° panoramic images and videos.

1.2.1 Inclusions

#	Feature	Description
1	<i>User Account Management</i>	Registration, login (email/phone/social), profile management, 2FA, password reset, and logout from all devices
2	<i>Panorama Upload Module</i>	Upload 360° panoramic images and videos from mobile device with compression options

3	<i>Panorama Management Module</i>	Organize, edit, rename, categorize, delete, and apply filters to panoramic content
4	<i>Interactive 360° Viewer</i>	Smooth viewer with touch gestures: swipe, zoom, rotate, and full-screen mode
5	<i>Virtual Tours & Storytelling</i>	Link multiple panoramas into sequential tours; add scene descriptions and navigation
6	<i>Interactive Hotspots Module</i>	Add clickable hotspots inside panoramas with text, audio, or video popups
7	<i>Social Interaction Module</i>	Likes, comments, sharing, follow/unfollow, real-time messaging, group chats, block users, star ratings
8	<i>Privacy & Access Control</i>	Public, private, or restricted visibility; content access permissions; account privacy settings
9	<i>Search & Discovery Module</i>	Search by keywords, categories, hashtags; browse trending content; basic recommendation engine
10	<i>Profile Insights Module</i>	View likes, comments, views, follower growth, and content popularity analytics
11	<i>Notification Module</i>	Push notifications for likes, comments, follows, messages, and platform announcements
12	<i>Admin & Moderation Panel</i>	Review reported content, manage user accounts, issue warnings or blocks, broadcast announcements
13	<i>UI/UX Design</i>	Intuitive mobile design, onboarding tour, smooth navigation, accessible layouts
14	<i>Testing & QA Reports</i>	Unit testing, integration testing, UAT results, bug reports, and performance metrics
15	<i>Deployment Package</i>	Final APK (Android) and IPA (iOS) ready for app store submission
16	<i>Complete Documentation</i>	SRS, user guide, technical documentation, and API documentation

1.2.2 Exclusions

Excluded Item	Reason
<i>Web-based version of the platform</i>	Current focus is solely on mobile application (Android and iOS)
<i>Advanced AI image processing (auto-stitching, OCR)</i>	Technically complex; deferred to version 2.0
<i>VR headset or professional 360° camera integration</i>	Requires specialized hardware beyond current project scope
<i>Payment gateway or subscription systems</i>	No monetization features in the academic version
<i>Enterprise-level cloud scaling / distributed infra</i>	Basic backend support is sufficient for academic project scale
<i>Legal / trademark registration services</i>	Not required for academic submission
<i>Automated content copyright detection</i>	Advanced AI feature; deferred to future version

1.3 Definitions, Acronyms, and Abbreviations

Term/Abbreviation	Definition
<i>360° Panorama</i>	An immersive image or video that captures a full spherical view of a scene, allowing users to look in all directions.
<i>Hotspot</i>	An interactive clickable element embedded within a panorama that displays additional content such as text, images, audio, or video when tapped.
<i>Virtual Tour</i>	A linked sequence of multiple panoramas that allows users to navigate between different scenes in a structured manner.
<i>Stitching</i>	The process of automatically aligning and combining multiple overlapping images into a single seamless panoramic image.
<i>FR</i>	Functional Requirement — a specification of what the system should do.
<i>NFR</i>	Non-Functional Requirement — a quality attribute the system must exhibit.

<i>QR</i>	Quality Requirement — a measurable performance or quality criterion.
<i>DESC</i>	Description — detailed explanation of a requirement.
<i>RAT</i>	Rationale — the reason why a requirement is necessary.
<i>DEP</i>	Dependency — other features or modules a requirement relies on.
<i>Admin / Administrator</i>	A user with elevated privileges responsible for managing the platform, users, and content.
<i>Registered User</i>	A user who has created an account and can upload panoramas, create tours, and interact with content.
<i>2FA</i>	Two-Factor Authentication — an additional security layer requiring a second verification step during login.
<i>UI</i>	User Interface — the visual elements through which users interact with the application.
<i>UX</i>	User Experience — the overall experience and satisfaction of a user when interacting with the application.
<i>API</i>	Application Programming Interface — a set of protocols for communication between frontend and backend.
<i>SDK</i>	Software Development Kit — a collection of tools for building applications (e.g., Flutter SDK).
<i>GPS</i>	Global Positioning System — used for geotagging panoramic content with location data.
<i>SSL/TLS</i>	Secure Sockets Layer / Transport Layer Security — encryption protocols for secure data transmission.
<i>MB</i>	Megabyte — a unit of digital information storage.
<i>MVP</i>	Minimum Viable Product — the version of the product with just enough features to be usable by early users.

1.4 References

- ♦ IEEE Software Engineering Standards Committee, "IEEE Std 830-1998, IEEE Recommended Practice for Software Requirements Specifications", October 20, 1998.

- ♦ Pressman, R.S., "Software Engineering: A Practitioner's Approach", 8th Edition, McGraw-Hill, 2014.

1.5 Assumptions & Dependencies

Dependency	Description	Impact if Unavailable
<i>Internet Connectivity</i>	Required for user authentication, content synchronization, media uploads/downloads, and all online features.	The application cannot access online services and most functionality becomes unavailable.
<i>GPS/Location Services</i>	Required for geotagging panoramas and location-based content discovery.	Users cannot add location data or view nearby panoramic content.
<i>Cloud Storage (AWS S3/Firebase Storage)</i>	Required for storing and retrieving panoramic images, videos, and virtual tour media files.	Media upload, storage, and download features will fail.
<i>MS SQL Server</i>	Required for storing user accounts, metadata, application settings, and system records.	The backend cannot perform data operations, resulting in complete system failure.
<i>Google Play Store / Apple App Store Approval</i>	Required for distributing the application to end users through official app marketplaces.	The application cannot be publicly released or downloaded by users.
<i>University Lab Resources</i>	Required for testing the application on multiple devices and network configurations.	Testing coverage becomes limited, increasing the risk of undetected issues.
<i>Flutter Framework</i>	Required for developing and building the cross-platform mobile application.	Mobile application development and deployment cannot proceed.
<i>Spring Boot Framework</i>	Required for developing the backend APIs and server-side functionality.	Backend services and API development cannot be implemented.

2 Feasibility Study

2.1 Technical Feasibility Study

The technical feasibility study evaluates whether the system can be built with available technologies, tools, and skills.

Feasibility Area	Assessment	Conclusion
<i>Mobile Development</i>	Flutter SDK supports Android and iOS development from a single codebase	Feasible
<i>Backend & APIs</i>	Spring Boot (Java) is a mature, widely-used backend framework	Feasible
<i>360° Panorama Viewing</i>	Established panorama libraries (e.g., Panorama360, custom WebGL viewers) are available	Feasible
<i>Database</i>	MS SQL Server Developer Edition is free; cloud alternatives available	Feasible
<i>Cloud Storage</i>	AWS S3 / Firebase Storage support large media files at affordable rates	Feasible
<i>Real-Time Messaging</i>	WebSocket or Firebase Realtime Database enables live chat features	Feasible
<i>AI Recommendations</i>	Deferred to v2.0; not required in MVP	Partially Deferred
<i>Security</i>	JWT authentication, encrypted storage, and HTTPS ensure data security	Feasible

2.2 Financial Feasibility Study

Financial feasibility evaluates whether the 3DPanoSphere mobile application can be developed, deployed, and maintained within reasonable financial limits. It analyzes the estimated development cost, funding possibilities, projected revenue opportunities, operational expenses, and long-term sustainability. The purpose of this analysis is to determine whether the expected benefits justify the required financial investment.

Financial Factor	Status
<i>Development Labor Cost</i>	PKR 0 (self-developed as academic project)
<i>Software Tools</i>	PKR 0 (all open-source: Flutter, Spring Boot, VS Code)
<i>Annual Cloud & Infrastructure</i>	~\$615/year (hosting, storage, domain)
<i>One-time Setup Costs</i>	~\$424 (camera optional, app store accounts)
<i>Contingency Budget</i>	\$346 (15% reserve)
<i>Funding Source</i>	Personal/academic funding
<i>Break-even Potential</i>	Achievable within the first year of commercialization

2.3 Operational Feasibility Study

The application is designed to be user-friendly and accessible on standard smartphones. No specialized hardware is required for end-users. The admin panel allows easy content moderation, and the system is designed to operate with minimal ongoing maintenance.

- ♦ Target users (real estate agents, tourists, students, general public) can adopt the app with minimal training
- ♦ Onboarding tour is built into the app to guide new users
- ♦ Backend infrastructure is manageable by a single developer during the academic phase
- ♦ Scalable architecture allows future growth without full redesign

2.4 Schedule Feasibility Study

Phase	Duration	Key Activities
<i>Requirement Gathering</i>	2 Weeks	Functional/non-functional requirements, feasibility studies
<i>System Design</i>	2 Weeks	Architecture, database design, UI/UX mockups
<i>Development</i>	4 Weeks	11 modules implemented (frontend + backend)
<i>Testing</i>	1 Week	Unit, integration, and user acceptance testing

Deployment	1 Week	App store submission, cloud setup, documentation
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2.5 Legal Feasibility Study

- ♦ User data is protected under applicable data privacy standards
- ♦ Content moderation tools are built in to handle copyright infringement and inappropriate content
- ♦ Users must accept Terms & Conditions and Privacy Policy before using the platform
- ♦ App store compliance guidelines (Google Play and Apple App Store) will be met before deployment

2.6 Social Feasibility Study

2.6.1 User Acceptance

The 3DPanoSphere platform is designed to provide an engaging and interactive experience for users who are interested in exploring and sharing panoramic content. Since social media and visual content platforms are widely used, users are likely to accept and adopt a system that allows them to explore immersive 360° images and videos.

2.6.2 Educational and Informational Value

The platform can provide educational benefits by allowing users to explore places virtually through panoramic images and virtual tours. Educational institutions, students, and researchers can use the system to explore historical sites, museums, and other locations without physically visiting them.

2.6.3 Support for Content Creators

3DPanoSphere can provide opportunities for photographers, travel bloggers, and digital creators to showcase their panoramic work. The platform allows them to share their content, gain recognition through ratings and interactions, and build a community around immersive visual media.

2.6.4 Community Engagement

The system encourages social interaction through features such as likes, comments, following other users, messaging, and star ratings. These features promote community engagement and collaboration among users with similar interests.

2.6.5 Social Impact

By providing a platform for immersive virtual exploration, the system can increase awareness of different locations, cultures, and environments. Virtual tours may also reduce the need for unnecessary travel in some cases, helping people explore places remotely.

3 Functional Requirements

3.1 User Class 1 - Registered User

3.1.1.1 Functional Requirement 2.1

ID: FR-RU-01

TITLE: User registration

DESC: Allow users to register using email, phone number, or social media accounts.

RAT: Provides secure entry to the platform.

DEP: User management module, authentication system

3.1.1.2 Functional Requirement 2.2

ID: FR-RU-02

TITLE: User login

DESC: Allow users to log in using registered credentials with two-factor authentication.

RAT: Ensures secure access and identity verification.

DEP: Authentication module

3.1.1.3 Functional Requirement 2.3

ID: FR-RU-03

TITLE: Password reset

DESC: Users can reset forgotten passwords via email or phone verification.

RAT: Supports account recovery in case of lost credentials.

DEP: User management module, email/SMS service

3.1.1.4 Functional Requirement 2.4

ID: FR-RU-04

TITLE: Logout from all devices

DESC: Users can log out remotely from all active sessions.

RAT: Enhances account security.

DEP: Session management module

3.1.1.5 Functional Requirement 2.5

ID: FR-RU-05

TITLE: Account deactivation/deletion

DESC: Users can deactivate or permanently delete their accounts.

RAT: Gives control over account lifecycle and privacy.

DEP: User management module

3.1.1.6 Functional Requirement 2.6

ID: FR-RU-06

TITLE: Profile creation

DESC: Users can create and fill personal information like name, bio, location, and contact info.

RAT: Personalizes user experience.

DEP: Profile database

3.1.1.7 Functional Requirement 2.7

ID: FR-RU-07

TITLE: Profile picture upload

DESC: Users can upload or update profile and cover images.

RAT: Visual identity for social interaction.

DEP: Media module

3.1.1.8 Functional Requirement 2.8

ID: FR-RU-08

TITLE: Privacy settings

DESC: Users can set account privacy (public/private) and manage post visibility.

RAT: Protects user content and personal data.

DEP: Privacy management module

3.1.1.9 Functional Requirement 2.9

ID: FR-RU-09

TITLE: Notification preferences

DESC: Users can configure alerts for comments, messages, and system notifications.

RAT: Enables personalized communication.

DEP: Notification module

3.1.1.10 Functional Requirement 2.10

ID: FR-RU-10

TITLE: Follow/unfollow users

DESC: Users can follow or unfollow other users to manage their feed.

RAT: Supports content discovery and social interaction.

DEP: User relationship module

3.1.1.11 Functional Requirement 2.11

ID: FR-RU-11

TITLE: Block/unblock users

DESC: Users can block or unblock accounts to control interactions.

RAT: Ensures a safe and comfortable environment.

DEP: User relationship module

3.1.1.12 Functional Requirement 2.12

ID: FR-RU-12

TITLE: Capture panorama

DESC: Users can capture multiple overlapping images for panoramic content.

RAT: Core feature for immersive visual storytelling.

DEP: Camera and media processing module

3.1.1.13 Functional Requirement 2.13

ID: FR-RU-14

TITLE: Panorama preview and save

DESC: Users can preview and save panoramic images to the gallery or device.

RAT: Allows users to review content before sharing.

DEP: Storage module

3.1.1.14 Functional Requirement 2.14

ID: FR-RU-15

TITLE: Panorama sharing

DESC: Users can share panoramic images directly to their profile or external platforms.

RAT: Supports social engagement and visibility.

DEP: Media sharing module

3.1.1.15 Functional Requirement 2.16

ID: FR-RU-16

TITLE: Upload image/video

DESC: Users can upload images or videos to their profile or galleries.

RAT: Allows content sharing and social engagement.

DEP: Media upload module

3.1.1.16 Functional Requirement 2.17

ID: FR-RU-17

TITLE: Apply filters & effects

DESC: Users can apply visual filters, effects, and edit uploaded media.

RAT: Enhances visual appeal and creativity.

DEP: Media editing module

3.1.1.17 Functional Requirement 2.18

ID: FR-RU-18

TITLE: Add captions & tags

DESC: Users can add descriptive captions, hashtags, or tags to their posts.

RAT: Improves content discoverability and context.

DEP: Content management module

3.1.1.18 Functional Requirement 2.19

ID: FR-RU-19

TITLE: Post visibility settings

DESC: Users can select whether a post is public, private, or visible to specific users.

RAT: Maintains user control over content audience.

DEP: Privacy module

3.1.1.19 Functional Requirement 2.20

ID: FR-RU-20

TITLE: Delete posts

DESC: Users can remove any post from their profile or gallery.

RAT: Ensures content management flexibility.

DEP: Content management module

3.1.1.20 Functional Requirement 2.21

ID: FR-RU-21

TITLE: Create time-limited stories

DESC: Users can create stories that expire after a set duration.

RAT: Encourages frequent engagement and content sharing.

DEP: Story module

3.1.1.21 Functional Requirement 2.22

ID: FR-RU-22

TITLE: Story customization

DESC: Add text, stickers, filters, or panoramic content to stories.

RAT: Enhances creativity and immersive experience.

DEP: Story editing module

3.1.1.22 Functional Requirement 2.23

ID: FR-RU-23

TITLE: Story highlights

DESC: Users can save selected stories permanently in highlights.

RAT: Allows long-term showcase of important content.

DEP: Story storage module

3.1.1.23 Functional Requirement 2.24

ID: FR-RU-24

TITLE: Story analytics

DESC: Users can view viewer count and interactions on their stories.

RAT: Provides insight into engagement and reach.

DEP: Analytics module

3.1.1.24 Functional Requirement 2.25

ID: FR-RU-25

TITLE: Delete story

DESC: Users can remove a story at any time before it expires.

RAT: Provides control over content lifecycle.

DEP: Story module

3.1.1.25 Functional Requirement 2.26

ID: FR-RU-26

TITLE: Create virtual gallery

DESC: Users can create dedicated gallery spaces to showcase their panoramic content.

RAT: Provides a structured platform for immersive visual storytelling.

DEP: Virtual exhibition module

3.1.1.26 Functional Requirement 2.27

ID: FR-RU-27

TITLE: Customize gallery layouts & themes

DESC: Users can personalize gallery design, layout, and theme.

RAT: Enhances user experience and branding.

DEP: Virtual exhibition module

3.1.1.27 Functional Requirement 2.28

ID: FR-RU-28

TITLE: Organize & accurate content

DESC: Users can sort, categorize, and arrange their uploaded content.

RAT: Improves content discoverability and user experience.

DEP: Gallery management module

3.1.1.28 Functional Requirement 2.29

ID: FR-RU-29

TITLE: Visitor analytics

DESC: Users can track visitor counts and interactions with their gallery.

RAT: Provides feedback on content popularity.

DEP: Analytics module

3.1.1.29 Functional Requirement 2.30

ID: FR-RU-30

TITLE: Send direct messages

DESC: Users can send text, audio, and video messages to other users.

RAT: Enables private communication and collaboration.

DEP: Messaging module

3.1.1.30 Functional Requirement 2.31

ID: FR-RU-31

TITLE: Create groups

DESC: Users can create groups for multiple users to communicate together.

RAT: Supports collaborative interactions and community building.

DEP: Messaging module

3.1.1.31 Functional Requirement 2.32

ID: FR-RU-32

TITLE: Add/remove members

DESC: Users can add or remove members from their groups.

RAT: Manages group participation and control.

DEP: Group management module

3.1.1.32 Functional Requirement 2.33

ID: FR-RU-33

TITLE: Pin/delete chats

DESC: Users can pin important chats or delete them from their inbox.

RAT: Enhances usability and chat management.

DEP: Messaging module

3.1.1.33 Functional Requirement 2.34

ID: FR-RU-34

TITLE: View post engagement metrics

DESC: Users can view likes, comments, and shares for their posts.

RAT: Provides insight into how well their content is performing.

DEP: Analytics module

3.1.1.34 Functional Requirement 2.35

ID: FR-RU-35

TITLE: Content performance analysis

DESC: Users can see detailed performance trends of their posts over time.

RAT: Helps users understand audience preferences and optimize content.

DEP: Analytics module

3.1.1.35 Functional Requirement 2.36

ID: FR-RU-36

TITLE: Best posting time recommendations

DESC: System suggests optimal posting times based on audience engagement patterns.

RAT: Improves content visibility and user interaction.

DEP: Analytics module

3.1.1.36 Functional Requirement 2.37

ID: FR-RU-37

TITLE: Profile visit analytics

DESC: Users can see the number of visitors to their profile.

RAT: Measures audience reach and popularity.

DEP: Analytics module

3.1.1.37 Functional Requirement 2.39

ID: FR-RU-39

TITLE: Audience growth tracking

DESC: Users can track the increase in followers over time.

RAT: Helps evaluate growth trends and social influence.

DEP: Analytics module

3.1.1.38 Functional Requirement 2.40

ID: FR-RU-40

TITLE: Push notifications for interactions

DESC: Users receive notifications for likes, comments, follows, and mentions.

RAT: Keeps users engaged and informed.

DEP: Notification module

3.1.1.39 Functional Requirement 2.41

ID: FR-RU-41

TITLE: Custom notification preferences

DESC: Users can enable or disable notifications for specific events.

RAT: Provides user control over notification preferences.

DEP: Notification module

3.1.1.40 Functional Requirement 2.43

ID: FR-RU-43

TITLE: Important updates & announcements

DESC: Users are informed of system updates, policy changes, or announcements.

RAT: Keeps the community informed and compliant with platform policies.

DEP: Notification module

3.1.1.41 Functional Requirement 2.44

ID: FR-RU-44

TITLE: Account privacy controls

DESC: Users can set account visibility to private or public.

RAT: Ensures user control over who can access their content.

DEP: Privacy module

3.1.1.42 Functional Requirement 2.45

ID: FR-RU-45

TITLE: Content visibility settings

DESC: Users can manage who can see individual posts.

RAT: Enhances content privacy and security.

DEP: Privacy module

3.1.1.43 Functional Requirement 2.46

ID: FR-RU-46

TITLE: Data sharing preferences

DESC: Users can control sharing of personal data with third-party apps.

RAT: Protects user data and ensures compliance with privacy policies.

DEP: Privacy module

3.1.1.44 Functional Requirement 2.49

ID: FR-RU-49

TITLE: Report inappropriate content

DESC: Users can report offensive or inappropriate posts and comments.

RAT: Helps maintain community safety and standards.

DEP: Moderation module

3.1.1.45 Functional Requirement 2.50

ID: FR-RU-50

TITLE: Appeal moderation decisions

DESC: Users can appeal if their content is flagged or removed.

RAT: Ensures fairness and accountability in content moderation.

DEP: Moderation module

3.1.1.46 Functional Requirement 2.51

ID: FR-RU-51

TITLE: View community guidelines

DESC: Users can access and read platform rules and guidelines.

RAT: Promotes awareness of acceptable behavior and content standards.

DEP: Moderation module

3.1.1.47 Functional Requirement 2.52

ID: FR-RU-52

TITLE: Create and manage virtual tours

DESC: Users can create virtual tours by linking multiple panoramas, adding scene descriptions, and enabling navigation between scenes.

RAT: Enables immersive storytelling and structured presentation of panoramic content.

DEP: Virtual Tours & Storytelling Module

3.1.1.48 Functional Requirement 2.53

ID: FR-RU-53

TITLE: Add interactive hotspots to panoramas

DESC: Users can add hotspots to panoramas showing additional text, images, audio, or video when clicked.

RAT: Enhances user engagement and content interactivity.

DEP: Interactive Features Module

3.1.1.49 Functional Requirement 2.54

ID: FR-RU-54

TITLE: Explore panoramas in full-screen interactive viewer

DESC: Users can zoom, rotate, and swipe panoramas smoothly for better viewing experience.

RAT: Provides immersive and intuitive panorama exploration.

DEP: Panorama Viewing Module

3.2 User Class 2 - Admin User

3.2.1.1 Functional Requirement 3.1

ID: FR-AU-1

TITLE: View all registered users

DESC: Admin can view a complete list of all registered users, including profile details and account status.

RAT: Enables monitoring and management of the platform's user base.

DEP: User management module

3.2.1.2 Functional Requirement 3.2

ID: FR-A-2U

TITLE: Block/unblock users

DESC: Admin can block users violating platform rules or unblock previously blocked users.

RAT: Ensures compliance with community guidelines and platform safety.

DEP: User management module

3.2.1.3 Functional Requirement 3.3

ID: FR-AU-3

TITLE: Delete user accounts

DESC: Admin can permanently remove user accounts that are inactive or violate rules.

RAT: Maintains platform integrity and reduces spam.

DEP: User management module

3.2.1.4 Functional Requirement 3.4

ID: FR-AU-4

TITLE: Verify user accounts

DESC: Admin can approve verification requests for users (email/phone/social media verification).

RAT: Confirms authenticity of user accounts and reduces fake profiles.

DEP: User management module

3.2.1.5 Functional Requirement 3.5

ID: FR-AU-5

TITLE: Review reported content

DESC: Admin can access all content flagged by users and review it for compliance.

RAT: Ensures timely removal of inappropriate material.

DEP: Moderation module

3.2.1.6 Functional Requirement 3.7

ID: FR-AU-7

TITLE: Enforce community guidelines

DESC: Admin can update and enforce rules regarding acceptable content and behavior.

RAT: Keeps the platform safe and socially responsible.

DEP: Moderation module

3.2.1.7 Functional Requirement 3.8

ID: FR-AU-8

TITLE: Manage appeals

DESC: Admin can review and act on user appeals regarding content removal or account actions.

RAT: Ensures fairness and transparency in moderation.

DEP: Moderation module

3.2.1.8 Functional Requirement 3.9

ID: FR-AU-9

TITLE: Platform usage analytics

DESC: Admin can view overall platform usage metrics such as daily active users, post engagement, and virtual gallery visits.

RAT: Helps in monitoring platform performance and growth trends.

DEP: Analytics module

3.2.1.9 Functional Requirement 3.10

ID: FR-AU-10

TITLE: Content engagement reports

DESC: Admin can see which posts, galleries, or stories are most engaged with by users.

RAT: Identifies trending content and informs moderation or feature improvements.

DEP: Analytics module

3.2.1.10 Functional Requirement 3.11

ID: FR-AU-11

TITLE: User activity reports

DESC: Admin can access reports of individual user activity, including login history and post interactions.

RAT: Helps detect suspicious or abusive behavior.

DEP: Analytics module

3.2.1.11 Functional Requirement 3.12

ID: FR-AU-12

TITLE: Update system configurations

DESC: Admin can modify platform settings such as notification preferences, security configurations, and feature toggles.

RAT: Ensures smooth operation and adaptability of the system.

DEP: System settings module

3.2.1.12 Functional Requirement 3.13

ID: FR-AU-13

TITLE: Manage content templates and filters

DESC: Admin can add or modify filters, categories, or templates for posts and virtual galleries.

RAT: Keeps content presentation consistent and user-friendly.

DEP: System settings module

3.2.1.13 Functional Requirement 3.14

ID: FR-AU-14

TITLE: System backup and restore

DESC: Admin can perform periodic backups and restore data in case of system failures.

RAT: Ensures data integrity and disaster recovery.

DEP: System settings module

3.2.1.14 Functional Requirement 3.15

ID: FR-AU-15

TITLE: Broadcast important updates

DESC: Admin can send platform-wide notifications, including announcements, maintenance alerts, or policy changes.

RAT: Keeps users informed and engaged with platform changes.

DEP: Notification module

3.2.1.15 Functional Requirement 3.16

ID: FR-AU-16

TITLE: Custom alerts for critical events

DESC: Admin receives notifications for urgent events such as security breaches, system errors, or mass reports of inappropriate content.

RAT: Enables rapid response and platform protection.

DEP: Notification module

4 Non-Functional Requirements

4.1 Performance

ID: NFR1

TAG: System Response Time

GIST: Response time of the system when users interact with panoramic content and galleries.

SCALE: The time required for loading panoramic images, user profiles, and virtual galleries.

METER: Measurements obtained from 1000 user requests during system testing.

MUST: The system should respond within 3 seconds.

ID: NFR2

TAG: Media Loading Performance

GIST: Performance of loading high-resolution panoramic images and media.

SCALE: The time required to load 360-degree panoramic images and videos.

METER: Measurements obtained from 500 media loading tests during testing.

MUST: Media content should load within 5 seconds.

ID: NFR3

TAG: Concurrent User Performance

GIST: System performance when multiple users access the platform simultaneously.

SCALE: Number of users that can use the system without performance degradation.

METER: Load testing with simulated concurrent users.

MUST: The system should support 500 concurrent users.

ID: NFR4

TAG: Data Processing Performance

GIST: Performance of processing user actions such as uploading panoramas, liking, commenting, and searching.

SCALE: Time required for processing user requests.

METER: Measurements obtained from 1000 system operations during testing.

MUST: Operations should be processed within 3 seconds.

ID: NFR5

TAG: Search Performance

GIST: Performance of the search feature when users search for galleries, users, or panoramic content.

SCALE: Time required for search results to appear.

METER: Measurements obtained from 1000 search queries during testing.

MUST: Search results should appear within 3 seconds.

4.2 Usability

ID: NFR6

TAG: System Usability

GIST: Ease of use of the 3DPanoSphere application for new and registered users.

SCALE: The ability of users to navigate the system, view panoramic galleries, and access features without confusion.

METER: Usability testing performed with 100 test users performing common tasks such as browsing galleries and viewing panoramas.

MUST: At least 85% of users should complete tasks successfully without assistance.

ID: NFR7

TAG: User Interface Clarity

GIST: Clarity and simplicity of the user interface.

SCALE: The interface should present clear navigation menus, icons, and instructions.

METER: Usability testing with 100 users evaluating interface clarity.

MUST: At least 85% of users should report the interface as easy to understand.

ID: NFR8

TAG: Navigation Efficiency

GIST: Efficiency of navigation within the application.

SCALE: Users should be able to reach major features such as gallery viewing, profile pages, and search within a small number of steps.

METER: Task completion testing with 100 users during usability evaluation.

MUST: Users should reach major features within 3 navigation steps.

ID: NFR9

TAG: User Learning Time

GIST: Time required for new users to understand and start using the system.

SCALE: The amount of time required for a new user to learn basic functions such as viewing panoramas and browsing galleries.

METER: Usability testing with new users during system trials.

MUST: New users should understand basic system functions within 10 minutes.

ID: NFR10

TAG: Accessibility Support

GIST: Accessibility support for different users.

SCALE: The application should support accessibility features such as readable text, clear icons, and responsive layouts.

METER: Accessibility evaluation based on standard usability guidelines.

MUST: The interface should support readable fonts and responsive design for different screen sizes.

4.3 Reliability

ID: NFR11

TAG: System Reliability

GIST: The reliability of the 3DPanoSphere system during normal operation.

SCALE: The system should correctly display panoramic images, user profiles, and galleries without system failure.

METER: Measurements obtained from 1000 system operations during testing.

MUST: More than 97% of operations should produce correct results without errors.

ID: NFR12

TAG: System Error Handling

GIST: The system's ability to handle errors and unexpected situations.

SCALE: The system should detect errors and display proper error messages without crashing.

METER: Testing with simulated system errors during 500 system operations.

MUST: The system should display proper error messages in 100% of detected errors.

ID: NFR13

TAG: Data Consistency

GIST: Consistency of user data and panoramic media in the system.

SCALE: User data such as profile information, uploaded panoramas, and gallery content should remain accurate and consistent across sessions.

METER: Measurements obtained from 1000 data update operations during testing.

MUST: Data consistency should be maintained in 98% of operations.

ID: NFR14

TAG: System Recovery

GIST: Ability of the system to recover after system failure or crash.

SCALE: The system should restore services and recover stored data after failure.

METER: Recovery tests performed after simulated system failures during testing.

MUST: System services should be restored within 5 minutes after failure.

4.4 Security

ID: NFR15

TAG: Communication Security

GIST: Security of communication between the user device and the system server.

SCALE: All login, registration, and data communication between the client and server should be encrypted.

METER: Security testing performed on 1000 login sessions during system testing.

MUST: 100% of login communication messages must be encrypted.

ID: NFR16

TAG: User Account Security

GIST: Security of registered user accounts.

SCALE: If a user attempts to log in with incorrect credentials, the system should deny access and notify the user.

METER: 1000 login attempts with invalid credentials during testing.

MUST: 100% of invalid login attempts should be rejected by the system.

ID: NFR17

TAG: Admin Account Security

GIST: Security of admin accounts.

SCALE: If an admin attempts to log in with incorrect credentials, the system must prevent login and display a login failure message.

METER: 1000 invalid admin login attempts during testing.

MUST: 100% of invalid login attempts must be blocked.

ID: NFR18

TAG: Account Lock Security

GIST: Protection against repeated failed login attempts.

SCALE: If a user or admin enters incorrect login credentials three consecutive times, the account should be temporarily locked.

METER: 1000 login attempts after account lock during testing.

MUST: The account should be locked for 30 minutes after three failed login attempts.

ID: NFR19

TAG: Data Privacy Security

GIST: Protection of user personal information stored in the system.

SCALE: Personal data such as email, password, and profile information must be securely stored and protected.

METER: Security testing of 1000 stored user records during testing.

MUST: 100% of sensitive user data must be securely stored using encryption.

ID: NFR20

TAG: Content Reporting Security

GIST: Protection of users from inappropriate or harmful content.

SCALE: Users should be able to report inappropriate content to the system administrators.

METER: Testing of 500 content reporting actions during system testing.

4.5 Scalability

ID: NFR21

TAG: System Scalability

GIST: The ability of the system to handle increasing numbers of users and content.

SCALE: The system should maintain stable performance when the number of users and panoramic media increases.

METER: Load testing with simulated concurrent users during system testing.

MUST: The system should support at least 500 concurrent users without performance degradation.

ID: NFR22

TAG: Media Storage Scalability

GIST: Scalability of storage for panoramic images and multimedia content.

SCALE: The system should store and manage increasing volumes of panoramic media without affecting performance.

METER: Storage testing with large volumes of panoramic images during testing.

MUST: The system should support storage of at least 10,000 panoramic media files.

ID: NFR23

TAG: Database Scalability

GIST: Scalability of the database when the number of users and data grows.

SCALE: The database should support increasing numbers of users, galleries, and media content.

METER: Database load testing with simulated user and content growth.

MUST: The database should support 10,000 registered users without performance degradation.

4.6 Maintainability

ID: NFR24

TITLE: Application maintainability

DESC: The application should be designed using modular architecture so developers can easily modify, update, or extend system components.

RAT: To allow efficient maintenance and future system upgrades.

DEP: None

ID: NFR25

TITLE: Application testability

DESC: Testing environments and debugging tools should be available for testing system functions such as login, gallery viewing, and media upload.

RAT: To ensure proper testing and error detection during development and maintenance.

DEP: None

ID: NFR26

TITLE: Code documentation

DESC: The system source code should include proper documentation and comments to support future development and maintenance.

RAT: To help developers understand the system structure and perform modifications easily.

DEP: None

4.7 Compliance

ID: NFR27

TITLE: Data privacy compliance

DESC: The system must protect user personal information such as email, profile details, and uploaded panoramic media according to standard data privacy regulations.

RAT: To ensure protection of user data and compliance with privacy policies.

DEP: Privacy policy and data protection mechanisms.

ID: NFR28

TITLE: Intellectual property compliance

DESC: Users retain ownership of the panoramic images, galleries, and other media they upload to the platform.

RAT: To comply with intellectual property laws and protect user content rights.

DEP: Terms and conditions module.

ID: NFR29

TITLE: Terms and conditions agreement

DESC: Users must accept the platform's terms and conditions before creating an account or uploading content.

RAT: To ensure users understand the platform policies and responsibilities.

DEP: User registration module.

4.8 Social & Ethical

ID: NFR30

TITLE: Community guidelines enforcement

DESC: The system should provide mechanisms for reporting inappropriate, offensive, or harmful content posted by users.

RAT: To maintain a safe and respectful community environment.

DEP: Content moderation system and admin panel.

ID: NFR31

TITLE: Prevention of harmful content

DESC: The platform should discourage posting of harmful, abusive, or misleading panoramic media and provide tools for moderation.

RAT: To ensure responsible and ethical use of the platform.

DEP: Moderation and reporting tools.

ID: NFR32

TITLE: User behavior monitoring

DESC: The system should monitor user activities to detect repeated violations of community guidelines.

RAT: To reduce harassment, spam, and misuse of the platform.

DEP: Moderation system and activity logs.

5 Prioritization Techniques

Prioritization techniques are used to determine which requirements should be implemented first based on factors such as business value, cost, risk, and stakeholder needs. For the 3DPanoSphere project, four prioritization techniques have been applied:

5.1 Moscow Method

MoSCoW is a prioritization technique that categorizes requirements into four groups. The name stands for Must have, Should have, Could have, and Won't have. This method helps stakeholders agree on what is essential for the current release and what can wait for future versions.

5.1.1 Categories

- ♦ **Must Have:** Critical requirements without which the system cannot function
- ♦ **Should Have:** Important requirements that add significant value but are not critical
- ♦ **Could Have:** Nice-to-have requirements that can be included if time permits
- ♦ **Won't Have:** Requirements explicitly excluded from the current release

Req No	Requirement	Priority (M/S/C/W)	Justification
01	<i>User registration (sign up)</i>	Must have	Essential for identifying and managing users
02	<i>User login and logout</i>	Must have	Required for secure access to personal accounts
03	<i>Profile creation (picture, bio, username)</i>	Must have	Basic user identity and personalization
04	<i>Profile editing and account updates</i>	Should have	Allows users to maintain up-to-date profiles
05	<i>Follow and unfollow other users</i>	Should have	Enhances social interaction but not core at launch
06	<i>Account verification</i>	Could have	Improves trust but optional for initial release
07	<i>Password reset</i>	Must have	Critical for account security and usability
08	<i>Block and unblock users</i>	Should have	Ensures user safety and privacy
09	<i>Upload 360° panoramic images</i>	Must have	Core app functionality
10	<i>Upload 360° panoramic videos</i>	Should have	Adds value but images are core
11	<i>Edit uploaded content (title, description, tags)</i>	Should have	Important for content organization
12	<i>Delete uploaded panoramas or videos</i>	Must have	Users need control over their content
13	<i>Organize content in galleries/albums</i>	Could have	Improves user experience, not essential at launch
14	<i>Add location/category to panoramas</i>	Could have	Helps discovery but optional initially
15	<i>Manage personal content library</i>	Should have	Enhances usability but not core

16	<i>Apply filters on panoramas</i>	Could have	Optional feature for better viewing experience
17	<i>Interactive 360° panorama viewer</i>	Must have	Core feature for exploring content
18	<i>Swipe, zoom, and rotate</i>	Must have	Essential for interactive viewing
19	<i>Full-screen viewing</i>	Should have	Enhances experience but not critical
20	<i>Smooth loading and navigation</i>	Must have	Key for usability and user retention
21	<i>View panoramas by others</i>	Must have	Central social interaction feature
22	<i>Create virtual tours using multiple panoramas</i>	Should have	Adds immersive experience, not core initially
23	<i>Link panoramas together</i>	Could have	Enhances navigation in tours
24	<i>Story mode for sequential scene viewing</i>	Could have	Optional for better storytelling
25	<i>Add descriptions to each scene</i>	Could have	Improves clarity but not critical
26	<i>Navigate between scenes</i>	Should have	Important for usability of tours
27	<i>Add interactive hotspots inside panoramas</i>	Should have	Enhances interactivity, optional initially
28	<i>Hotspots showing text info</i>	Could have	Optional enhancement for user experience
29	<i>Hotspots with audio/video</i>	Could have	Nice to have for richer content
30	<i>Clickable objects for exploration</i>	Could have	Optional for advanced interactivity
31	<i>Highlight important locations</i>	Should have	Improves content navigation
32	<i>Like panoramas and videos</i>	Must have	Core social feature

33	<i>Comment on posts</i>	Must have	Core social interaction
34	<i>Share panoramas</i>	Should have	Enhances content visibility
35	<i>Follow and interact with creators</i>	Should have	Important for engagement
36	<i>Real-time messaging / chat</i>	Could have	Nice-to-have for socialization
37	<i>Pin important chats</i>	Won't have	Not needed in initial release
38	<i>Create chat groups</i>	Won't have	Not needed in initial release
39	<i>All users can send messages in a group</i>	Won't have	Not needed in initial release
40	<i>Delete chats</i>	Should have	Important for privacy
41	<i>View number of likes and comments</i>	Should have	Helps users track engagement
42	<i>Track panorama views and engagement</i>	Could have	Optional analytics for users
43	<i>View followers growth</i>	Could have	Optional for social insights
45	<i>Search panoramas or users using keywords</i>	Must have	Core for finding content
46	<i>Explore trending/popular content</i>	Should have	Enhances discovery experience
48	<i>Discover new creators and tours</i>	Could have	Nice to have for social exploration
49	<i>Filter content by popularity/views</i>	Could have	Optional enhancement
51	<i>Follower notifications</i>	Could have	Optional at launch
52	<i>Message notifications</i>	Could have	Optional at launch
53	<i>Platform updates</i>	Should have	Important for user awareness

54	<i>Manage account settings</i>	Must have	Essential for basic account control
55	<i>Public/private profile</i>	Should have	Provides privacy options
56	<i>Control visibility/comments</i>	Should have	Enhances privacy and safety
57	<i>Manage blocked users</i>	Could have	Optional at initial release
58	<i>Notification preferences</i>	Could have	Optional customization
59	<i>Monitor platform</i>	Must have	Critical for system integrity
60	<i>Review reports</i>	Must have	Essential for moderation
61	<i>Remove posts</i>	Must have	Ensures platform safety
62	<i>Manage users</i>	Must have	Admin core functionality
63	<i>Announcements</i>	Should have	Important for communication
64	<i>Security & moderation</i>	Must have	Ensures platform stability

5.2 100-Points Method

The 100-Points Method is a stakeholder-driven prioritization technique used to determine the relative importance of system features. In this approach, each stakeholder is allocated 100 points and asked to distribute them among the proposed features according to their perceived value and importance. Features that receive a higher number of points are considered higher priority, while those receiving fewer points are considered less critical.

Process:

- ♦ Each stakeholder receives a total of 100 points.
- ♦ Points are distributed among the available features based on importance.
- ♦ Critical features receive more points, while less important features receive fewer points.
- ♦ The points from all stakeholders are aggregated.

- ♦ Features with higher total scores are assigned higher priority in the development process.

Functionality	Marketing Representative	IT Manager	Business Head	Moscow Priority
<i>User Registration & Login</i>	15	20	15	Must Have
<i>Profile Management</i>	10	10	10	Should have
<i>Upload & Manage 360° Panoramas</i>	25	25	25	Must Have
<i>Interactive 360° Panorama Viewer</i>	20	15	15	Must Have
<i>Virtual Tours & Storytelling</i>	5	5	5	Could Have
<i>Social Interaction</i>	10	5	10	Must Have
<i>Search & Discovery</i>	5	5	5	Should Have
<i>Notifications & Settings</i>	5	0	5	Could Have
<i>Admin & Moderation</i>	5	15	10	Must Have

Total Points: 100

5.3 Value-Based Requirements

Functionality	Business Value	Implementation Cost	Value/Cost Ratio
<i>User Registration & Login</i>	10	4	2.50
<i>Upload & Manage 360° Panoramas</i>	10	6	1.67
<i>Interactive 360° Panorama Viewer</i>	9	6	1.50

<i>Admin & Moderation</i>	8	6	1.33
<i>Profile Management</i>	8	7	1.14
<i>Social Interaction</i>	7	6	1.17
<i>Search & Discovery</i>	8	5	1.60
<i>Notifications & Settings</i>	5	5	1.00
<i>Virtual Tours & Storytelling</i>	6	8	0.75

Higher Ratio = Higher Priority (more value, less cost)

Lower Ratio = Lower Priority (more costly, less benefit)

5.4 Analytic Hierarchy Process

The Analytic Hierarchy Process (AHP) is used to analyze and prioritize complex decisions by breaking them down into smaller, more manageable components. The process begins by defining the main goal, followed by identifying the criteria that influence the decision. Alternatives are then evaluated through pairwise comparisons against each criterion to determine their relative importance. AHP is particularly useful when decision-making involves multiple factors such as cost, performance, security, usability, and reliability, allowing decision-makers to reach a well-structured and justified conclusion.

5.4.1 Goal

To select the best system deployment option for the 3DPanoSphere project

5.4.2 Alternatives

The following three alternatives are considered:

- ♦ Cloud-Based System
- ♦ Hybrid System
- ♦ On-Premises System

5.4.3 Criteria

The decision is based on the following criteria:

Criteria	Description
----------	-------------

Cost	Initial and maintenance cost of the system
Performance	Speed and responsiveness of the application
Scalability	Ability to handle increasing users/data
Reliability	System stability and uptime
Security	Protection of user data and system

5.4.4 Criteria Weights

Each criterion is assigned a weight based on its importance:

Criteria	Weight
Cost	0.10
Performance	0.20
Scalability	0.25
Reliability	0.20
Security	0.25

5.4.5 Evaluation of Alternatives

1. Cost

Option	Weight
Cloud	0.5
Hybrid	0.3
On-Premises	0.2

2. Performance

Option	Weight
Cloud	0.3
Hybrid	0.4
On-Premises	0.3

3. Scalability

Option	Weight
Cloud	0.5
Hybrid	0.3
On-Premises	0.2

4. Reliability

Option	Weight
Cloud	0.4
Hybrid	0.3
On-Premises	0.3

5. Security

Option	Weight
Cloud	0.3
Hybrid	0.4
On-Premises	0.3

5.4.6 Final Score Calculation

The final score is calculated using:

$$\text{Final Score} = \Sigma (\text{Criteria Weight} \times \text{Option Weight})$$

1. Cloud-Based System

$$\begin{aligned} &= (0.10 \times 0.5) + (0.20 \times 0.3) + (0.25 \times 0.5) + (0.20 \times 0.4) + (0.25 \times 0.3) \\ &= 0.05 + 0.06 + 0.125 + 0.08 + 0.075 \\ &= \mathbf{0.39} \end{aligned}$$

2. Hybrid System

$$\begin{aligned} &= (0.10 \times 0.3) + (0.20 \times 0.4) + (0.25 \times 0.3) + (0.20 \times 0.3) + (0.25 \times 0.4) \\ &= 0.03 + 0.08 + 0.075 + 0.06 + 0.10 \\ &= \mathbf{0.345} \end{aligned}$$

3. On-Premises System

$$= (0.10 \times 0.2) + (0.20 \times 0.3) + (0.25 \times 0.2) + (0.20 \times 0.3) + (0.25 \times 0.3)$$

$$= 0.02 + 0.06 + 0.05 + 0.06 + 0.075$$

$$= \mathbf{0.265}$$

5.4.7 Final Observations

Option	Score
Cloud	0.39
Hybrid	0.345
On-Premises	0.265

6 UML Modeling

6.1 Use Case Diagram for Registered User

The Registered User interacts with the 3DPanoSphere mobile application to view, upload, and interact with panoramic content. The user can also follow other users, search for content, create virtual tours, and communicate with other users through messaging.

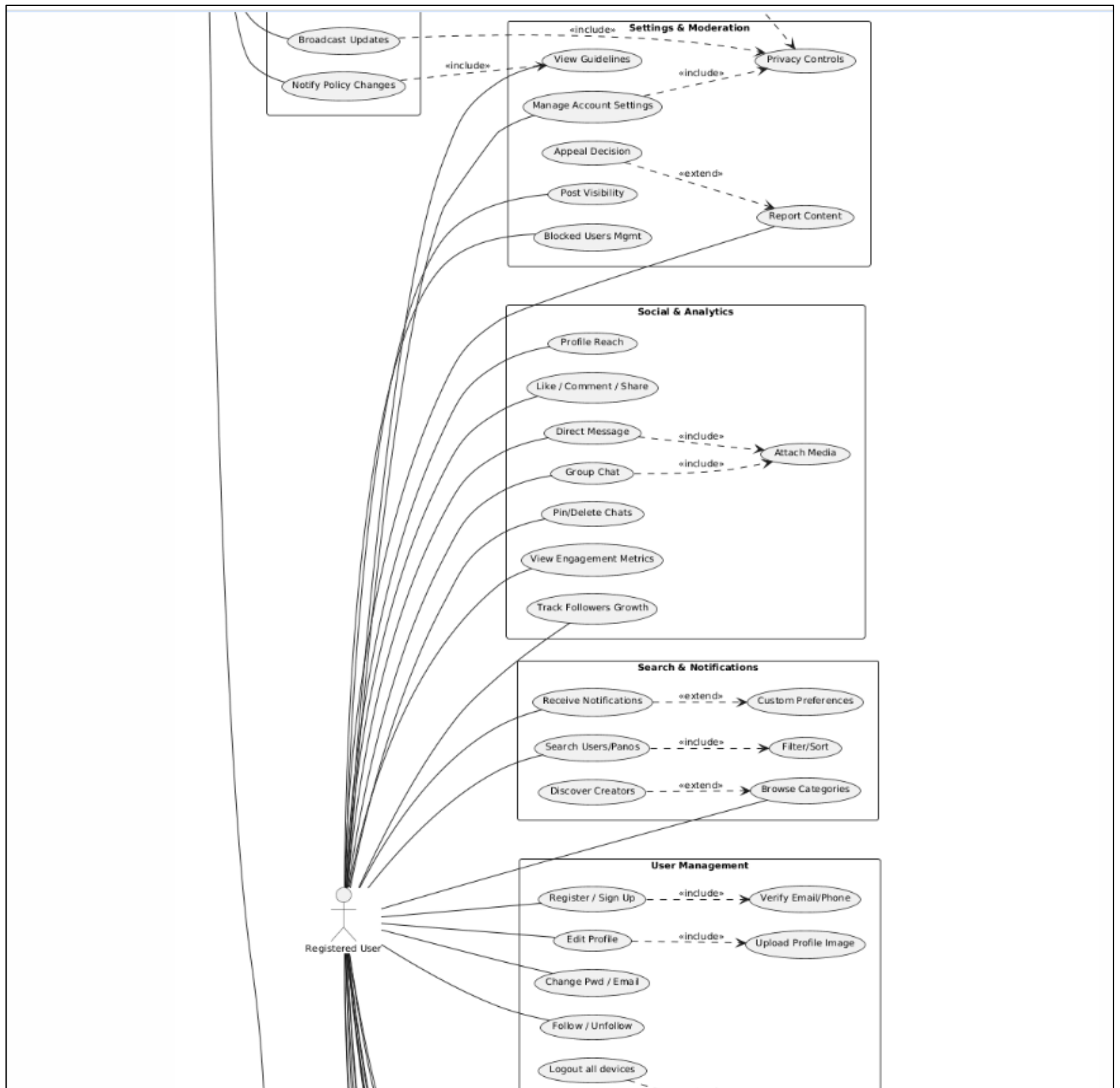


Figure 1: Use Case Diagram 01 for Registered User

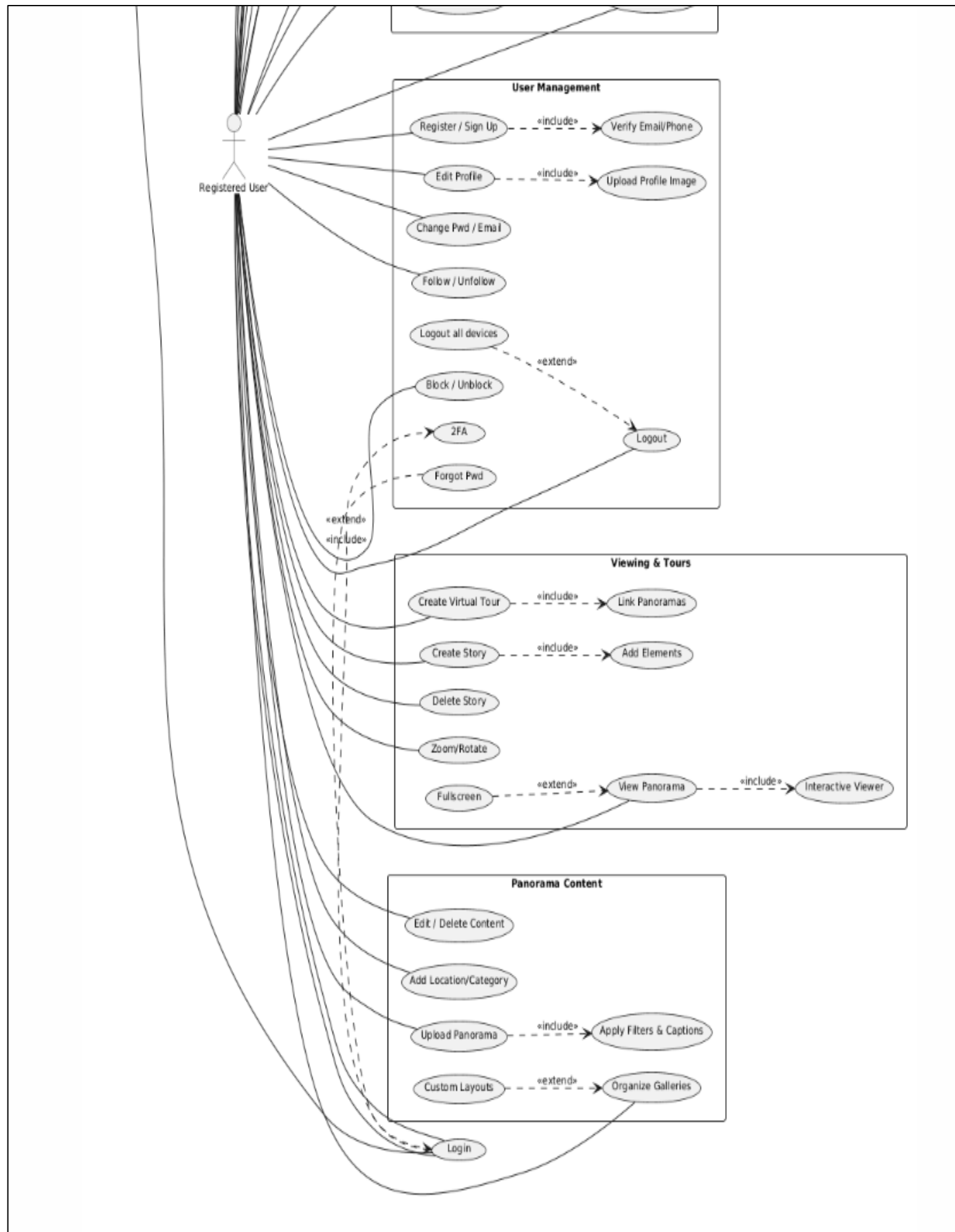


Figure 2: Use Case Diagram 02 for Registered User

6.2 Use Case Diagram for Admin User

The Admin User manages and monitors the 3DPanoSphere platform. Admin responsibilities include user management, content moderation, reviewing reported content, issuing warnings or blocks to violating users, managing user appeals, viewing platform analytics, and broadcasting announcements to all users. The admin ensures platform safety and compliance with community guidelines.

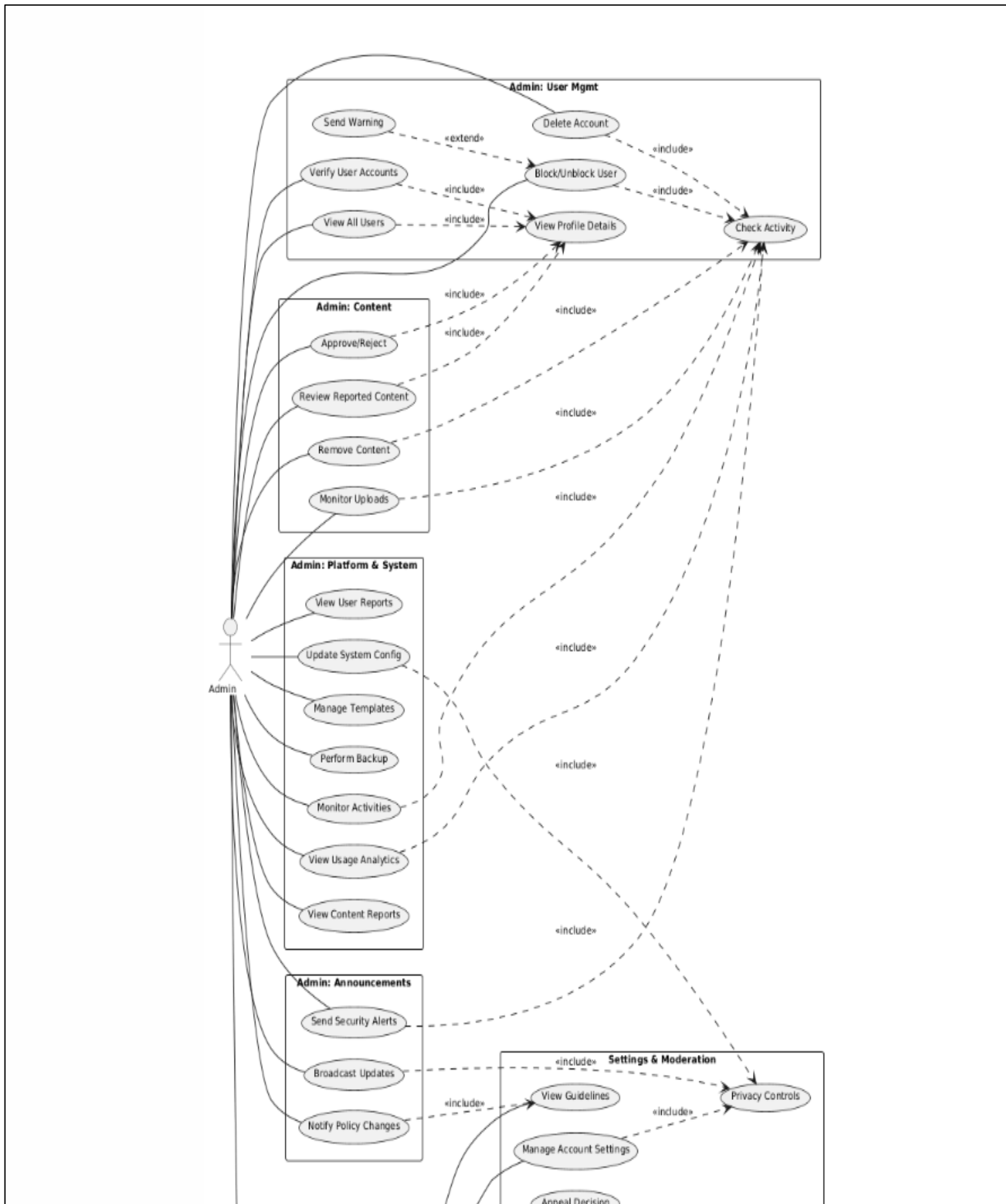
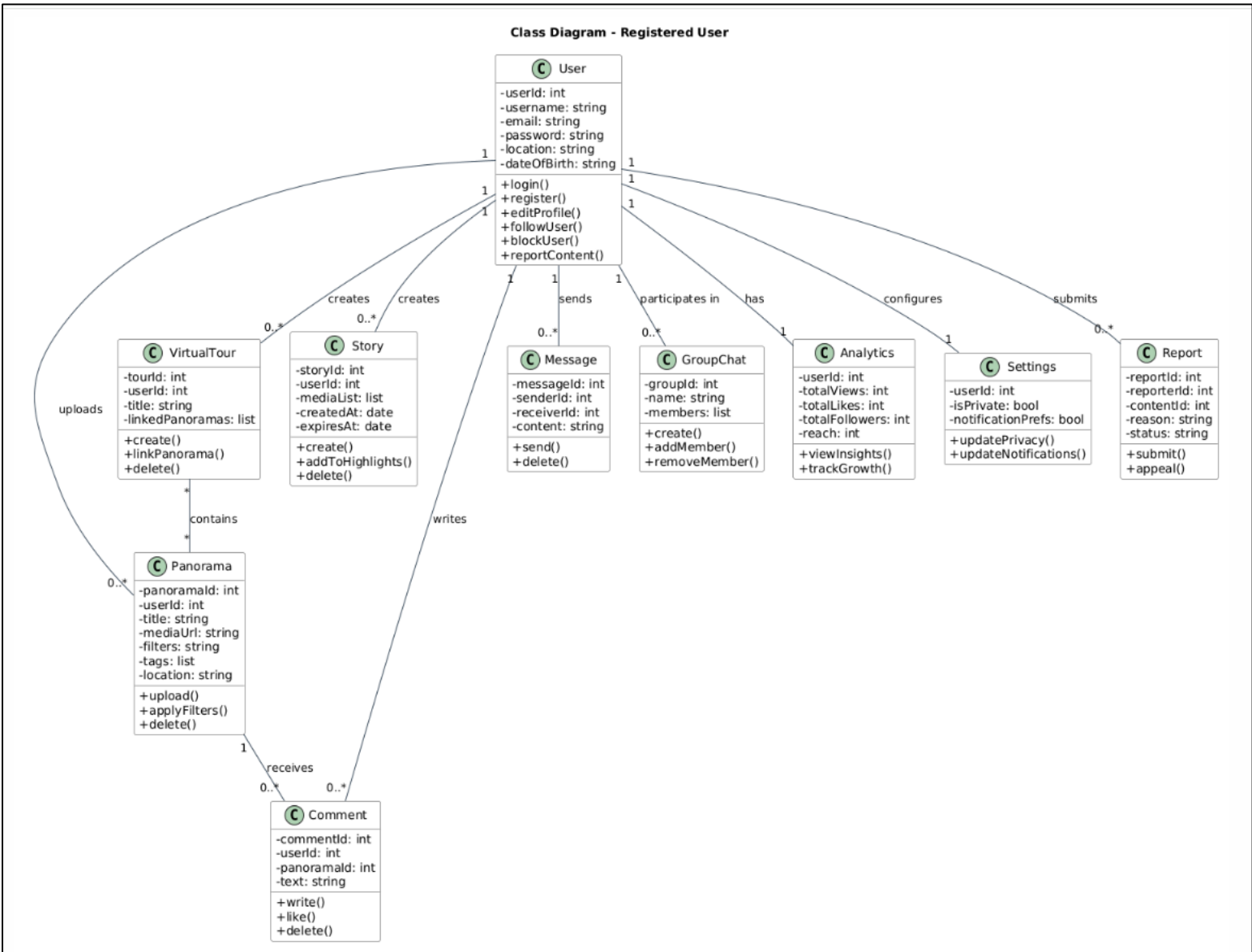


Figure3: Use Case Diagram for Admin User

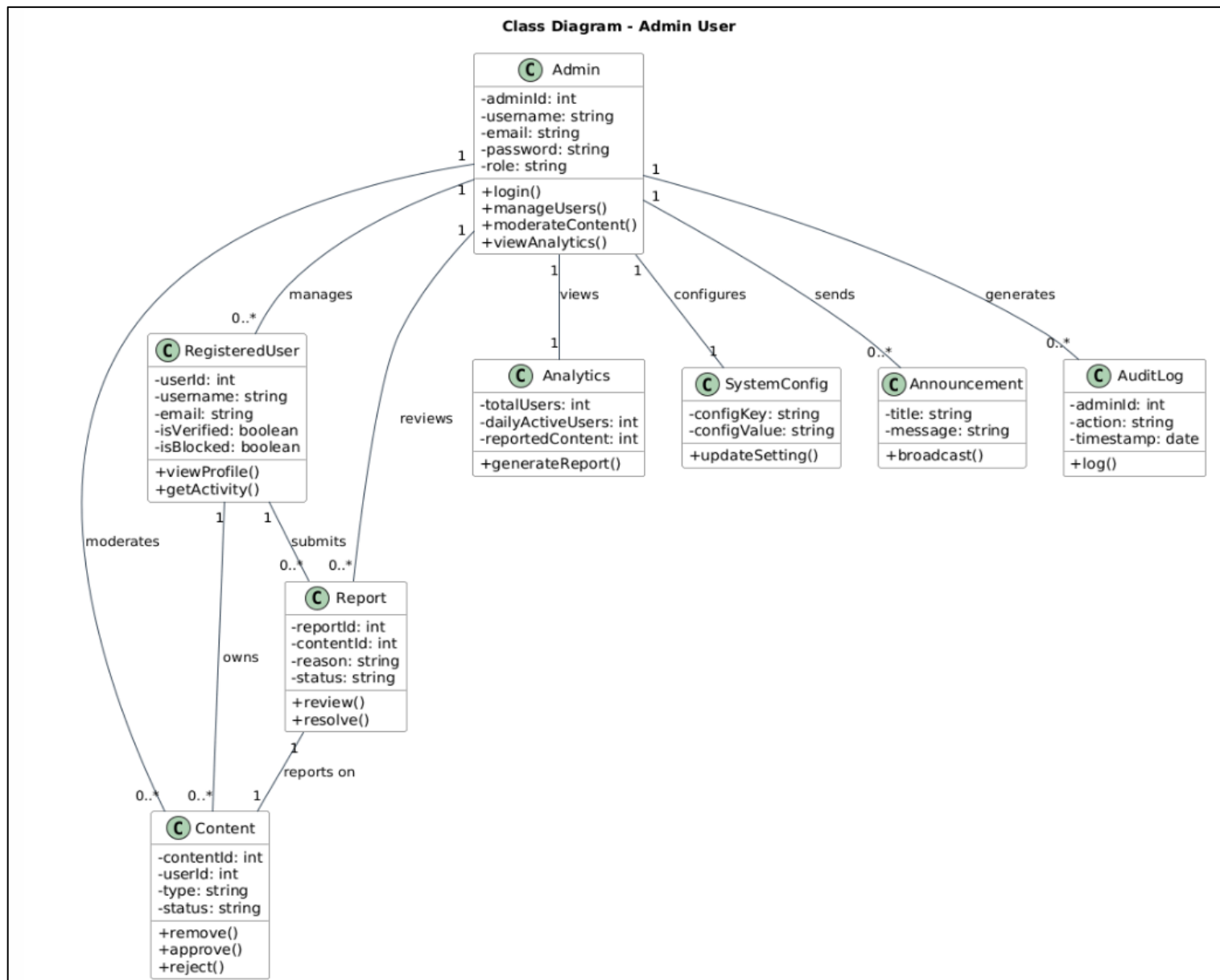
6.3 Class Diagram for Registered User

The Registered User class diagram represents all features available to a normal user of the 3DPanoSphere platform. The main class is **User** which handles authentication, profile management, follow/block actions, and content reporting. Users can upload **Panorama** (360° images/videos) with filters and tags, create **Virtual Tours** by linking multiple panoramas, and post temporary **Stories** that expire after a set time. For social interaction, users can write **Comments** on panoramas, send private **Messages**, and participate in **Group Chats**. Each user has **Analytics** to track views, likes, followers, and reach, plus **Settings** to manage privacy and notification preferences. Users can also submit **Reports** for inappropriate content and appeal moderation decisions.



6.4 Class diagram for Admin User

The Admin User class diagram represents all platform management features available to an administrator. The main class is **Admin** which handles login, user management, content moderation, and analytics viewing. Admins can manage **RegisteredUsers** by blocking, unblocking, deleting, or verifying accounts. They moderate user **Content** by reviewing, approving, rejecting, or removing inappropriate posts. Admins also review user-submitted **Reports** and resolve them by taking necessary actions. The **Analytics** class provides platform-wide statistics like total users, daily active users, and reported content counts. Admins can update **SystemConfig** settings, broadcast **Announcements** to all users, and all actions are logged in **AuditLog** for security and accountability.



6.5 Activity Diagrams for Admin Users

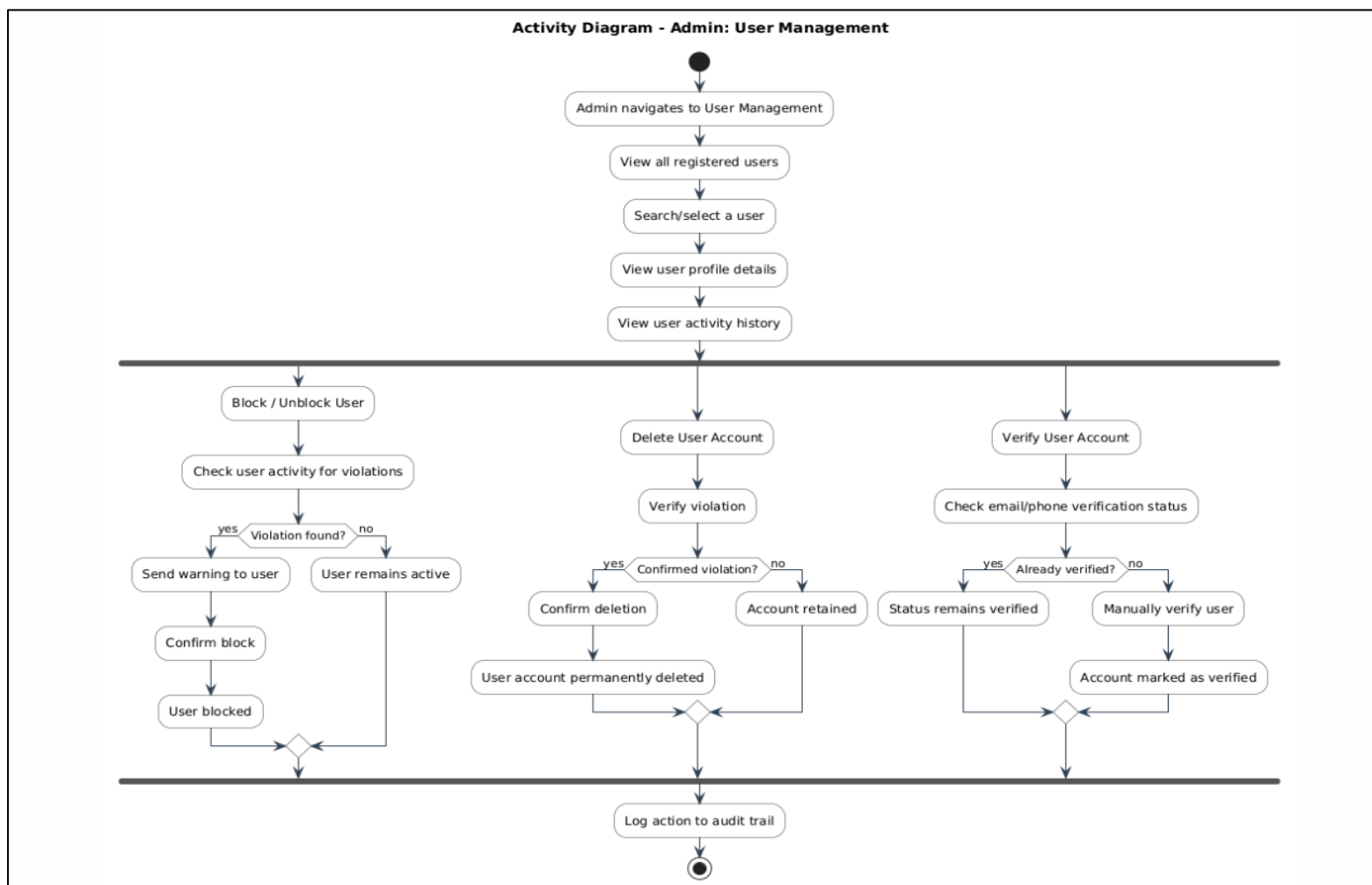


Figure 4: Class Diagram for Admin – User Management

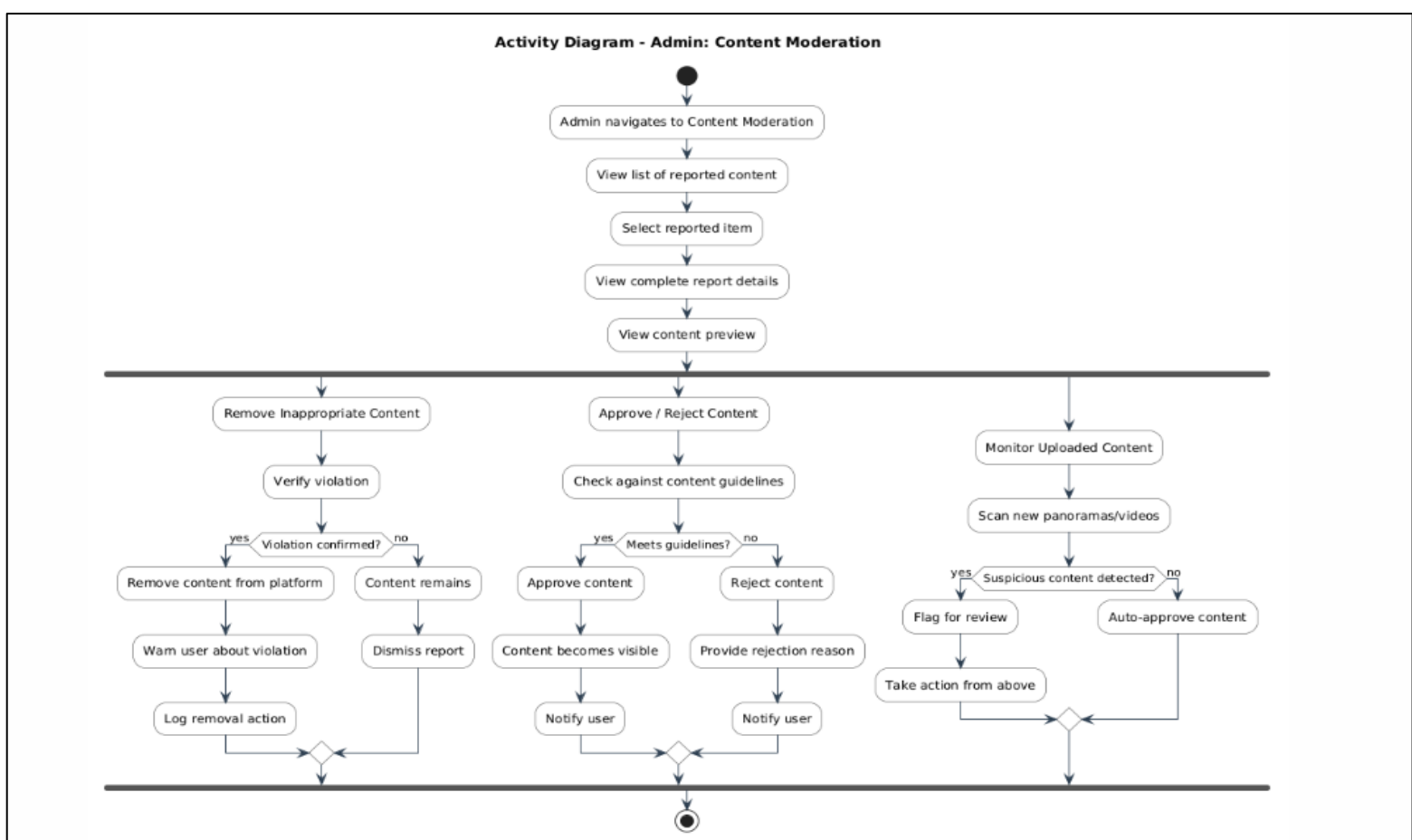


Figure 5: Class Diagram for Admin – Content Moderation

6.6 Activity Diagrams for Registered Users

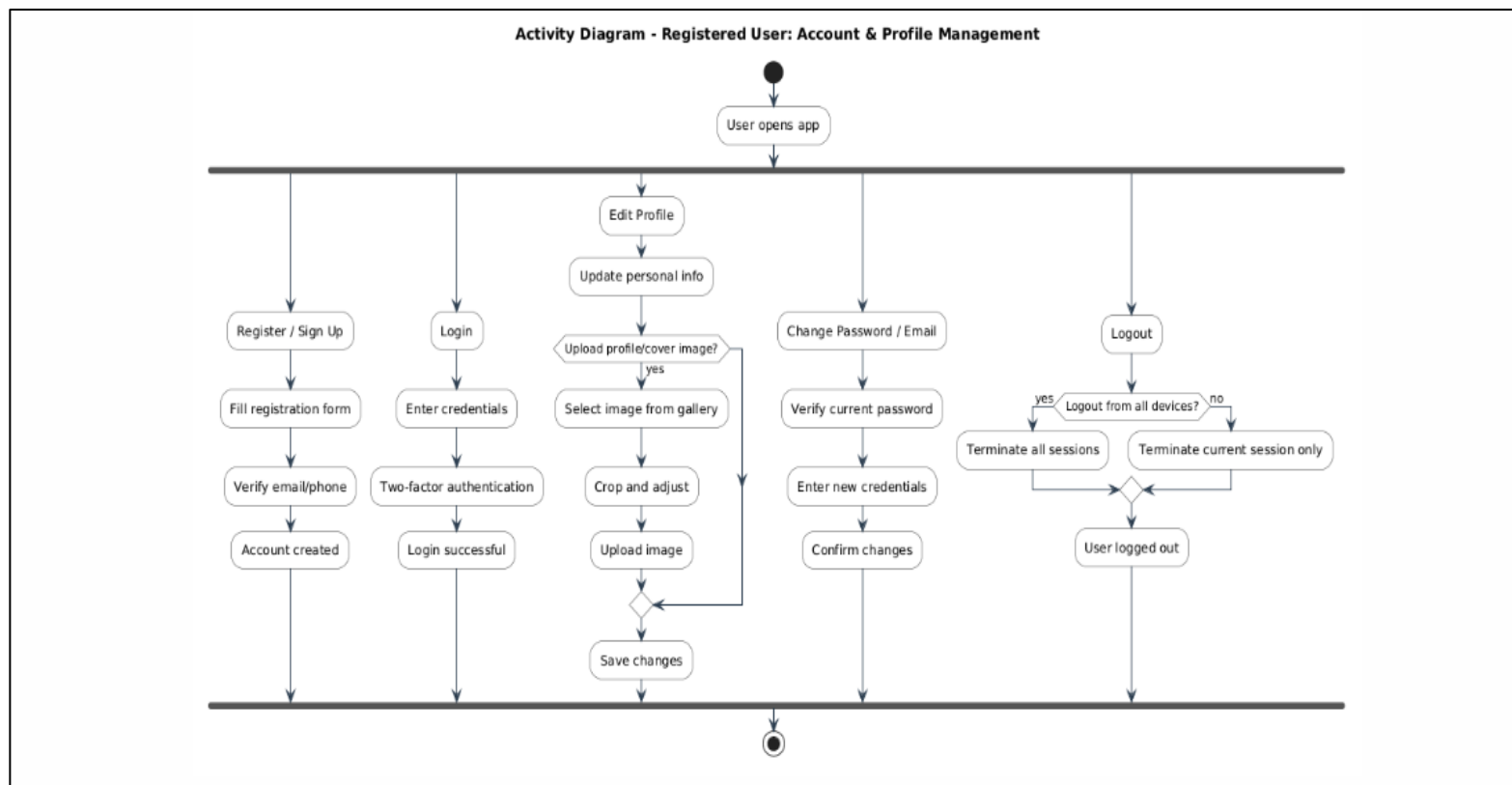


Figure 6: Class Diagram for User – User Account & Profile Management

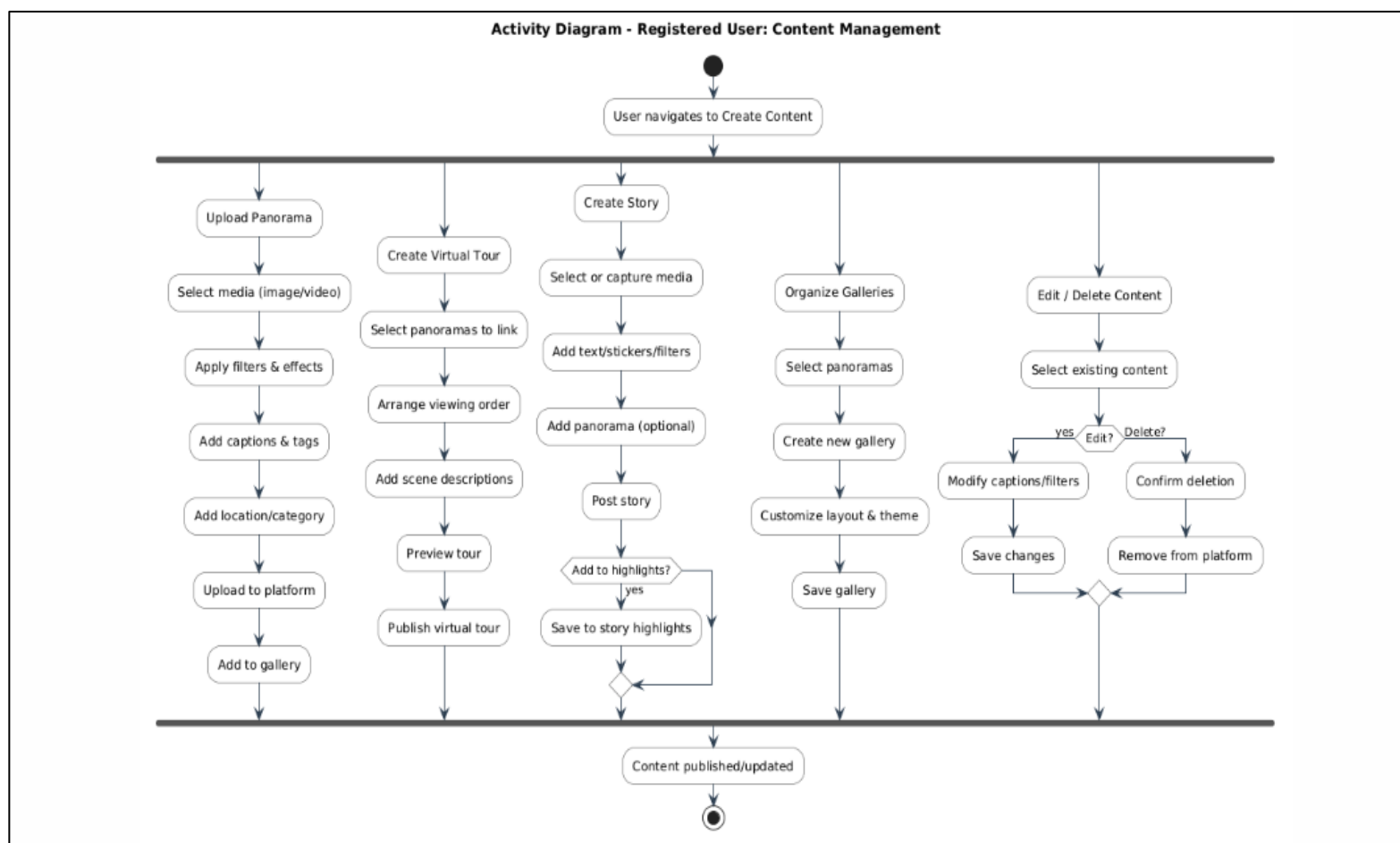
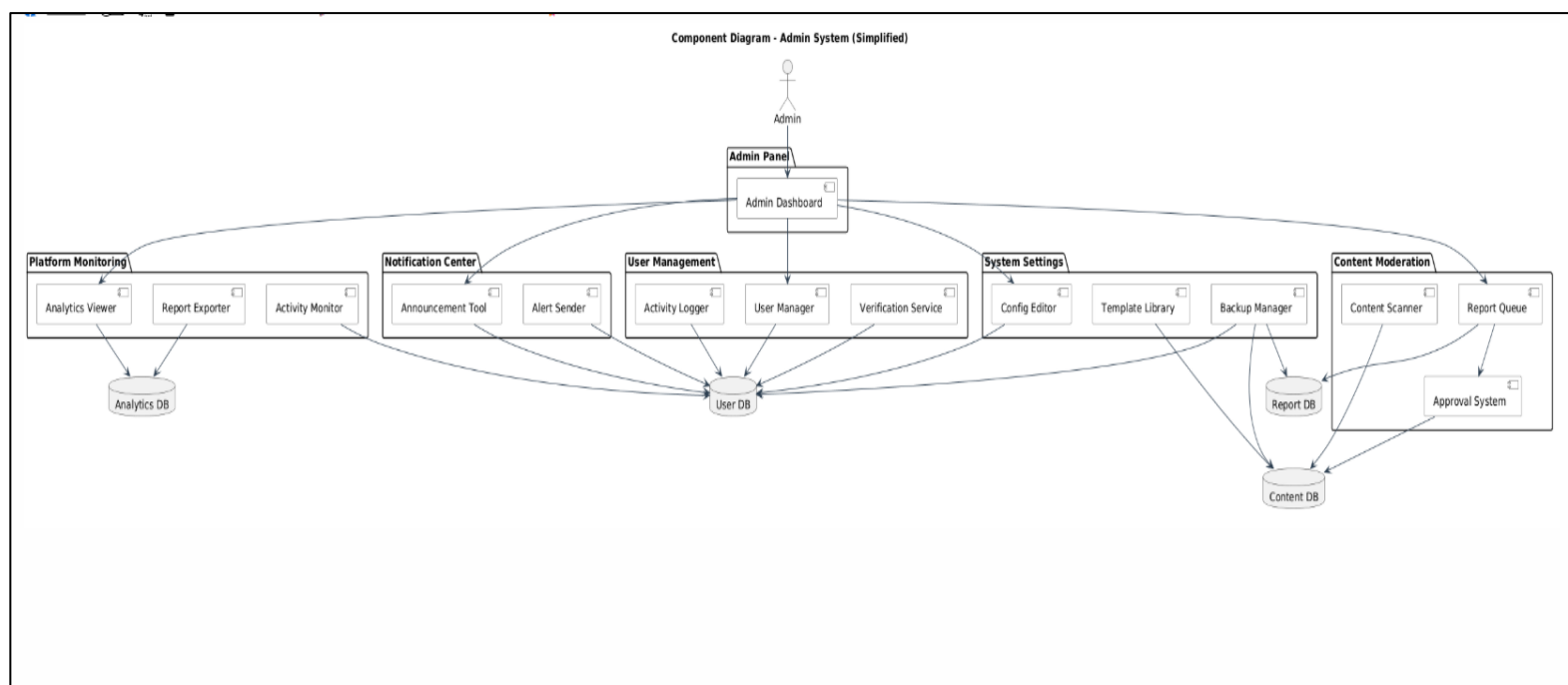
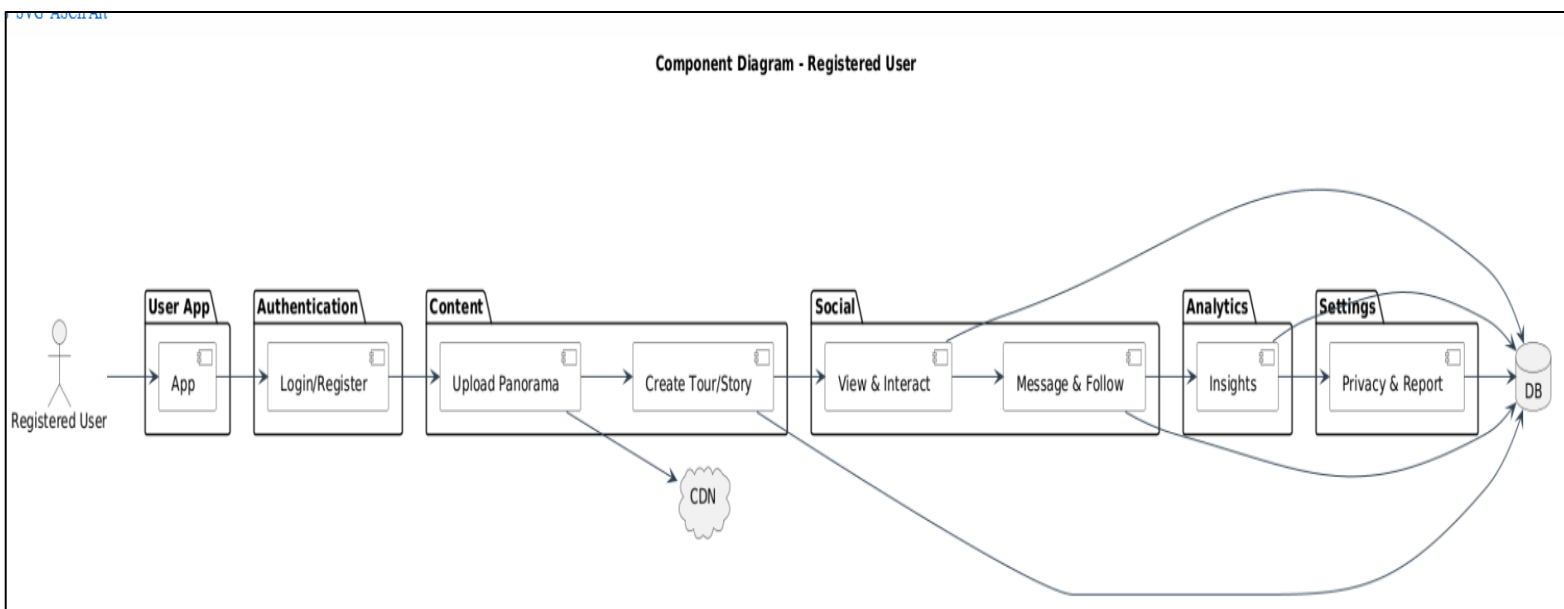


Figure 7: Class Diagram for User – Content Management

6.7 Component Diagram for Amin User



6.8 Component Diagram for Registered User



7 Data Flow Diagram (DFD)

7.1 Level 0 DFD

External Entities:

1. Registered User
2. Admin

System:

- 3DPanoSphere System

Data Flows:

1. From Registered User to System:

- Registration Data
- Login Credentials
- Panorama / Story / Tour
- Comments / Likes / Messages

2. From System to Registered User:

- Verification Code
- User Profile
- Content Feed
- Notifications
- Analytics

3. From Admin to System:

- Login Credentials
- Moderation Actions
- System Config
- Announcements

4. From System to Admin:

- User List
- Reported Content
- Analytics Reports
- Audit Logs

5. From System to Database:

- Read/Write Data

6. From Database to System:

- Retrieved Data

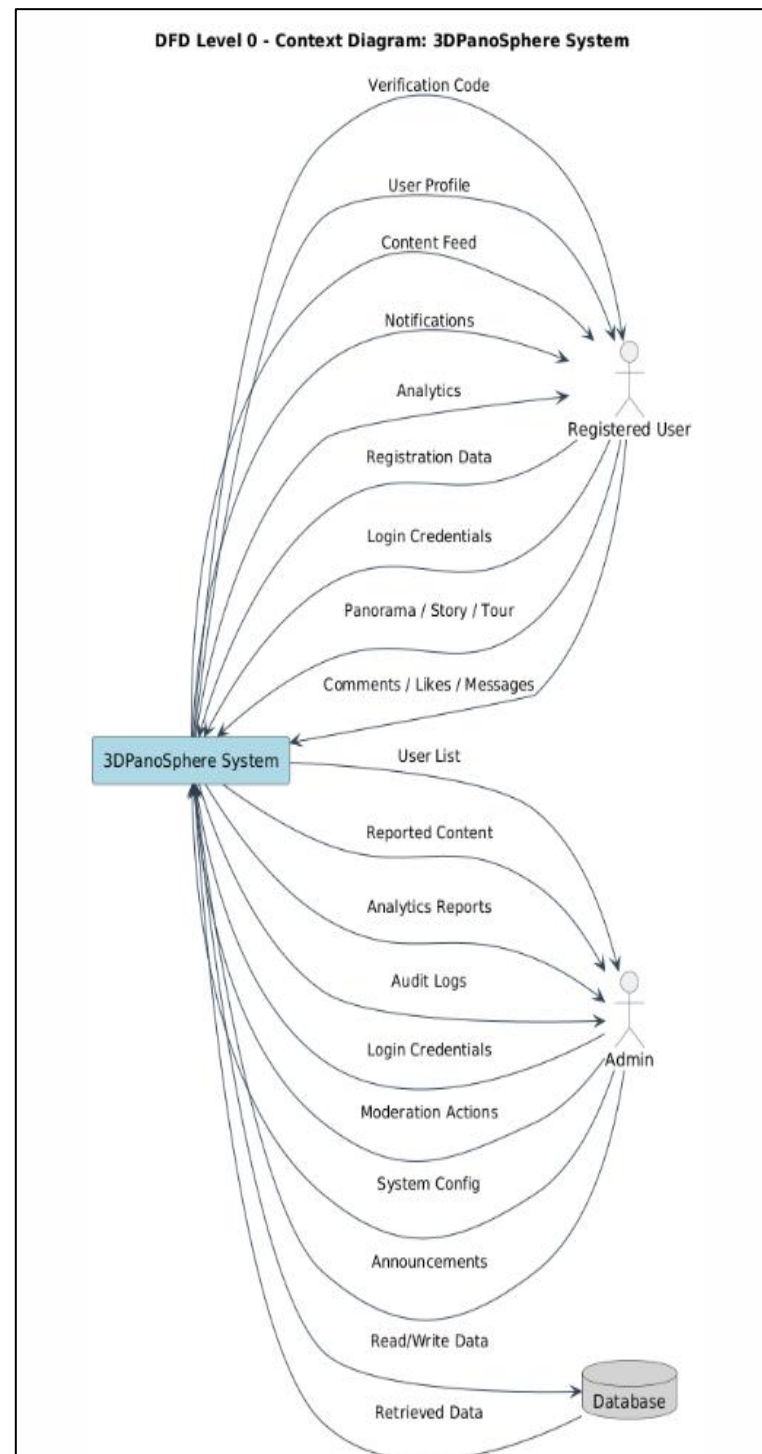


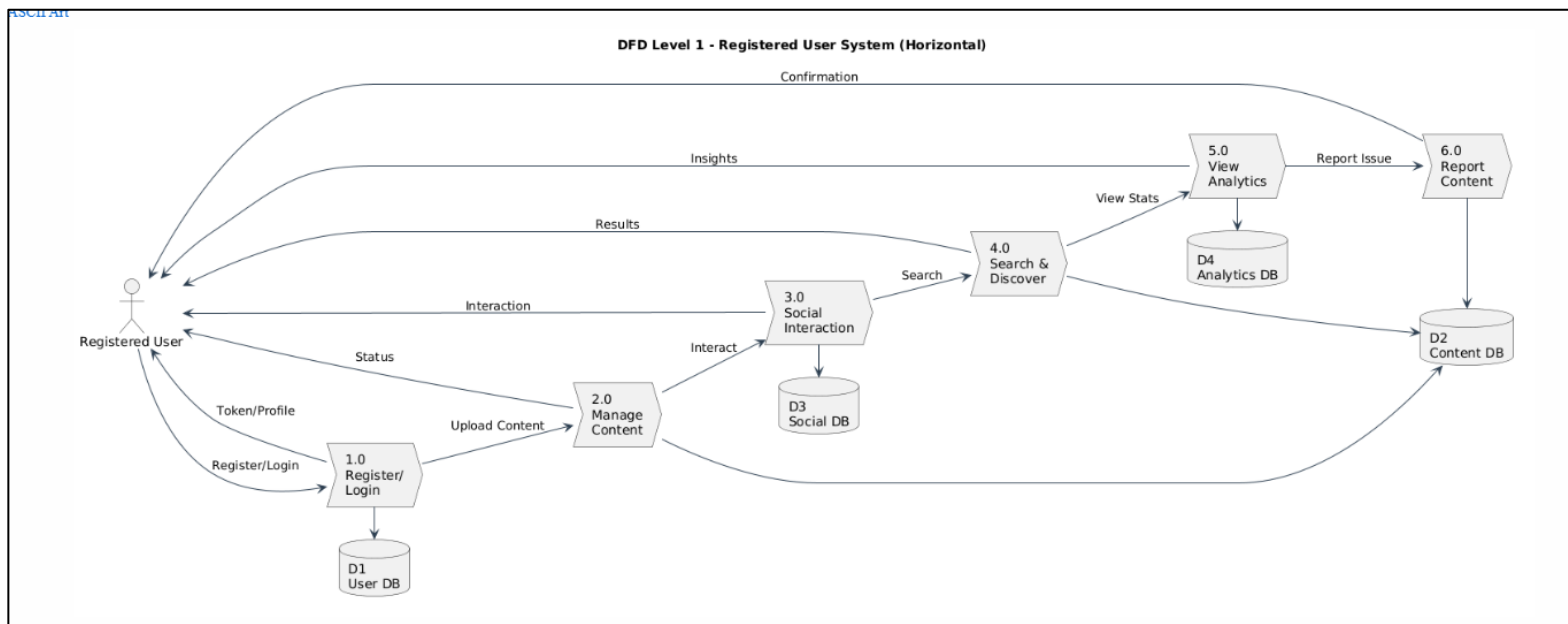
Figure 1: DFD Level 0 Diagram

7.2 Level 1 DFD - Registered User Subsystem

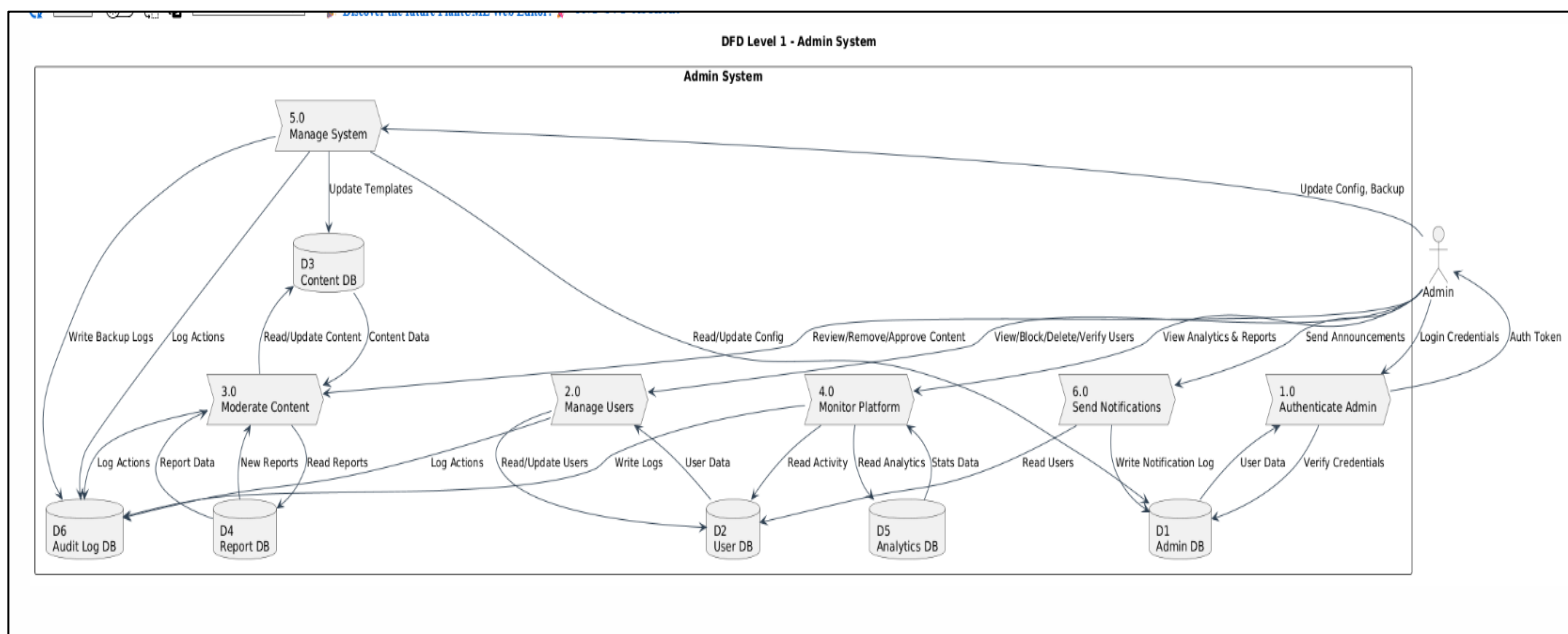
The Level 1 DFD for Registered User has 6 processes.

- ♦ **Process 1.0 (Register/Login)** handles authentication and profile management using User DB.
- ♦ **Process 2.0 (Manage Content)** allows uploading panoramas, tours, and stories to Content DB.
- ♦ **Process 3.0 (Social Interaction)** manages likes, comments, messages, and follows using Social DB.
- ♦ **Process 4.0 (Search & Discover)** searches users and content from Content DB and User DB.
- ♦ **Process 5.0 (View Analytics)** provides engagement insights from Analytics DB.
- ♦ **Process 6.0 (Report Content)** handles content reporting and updates flags in Content DB.

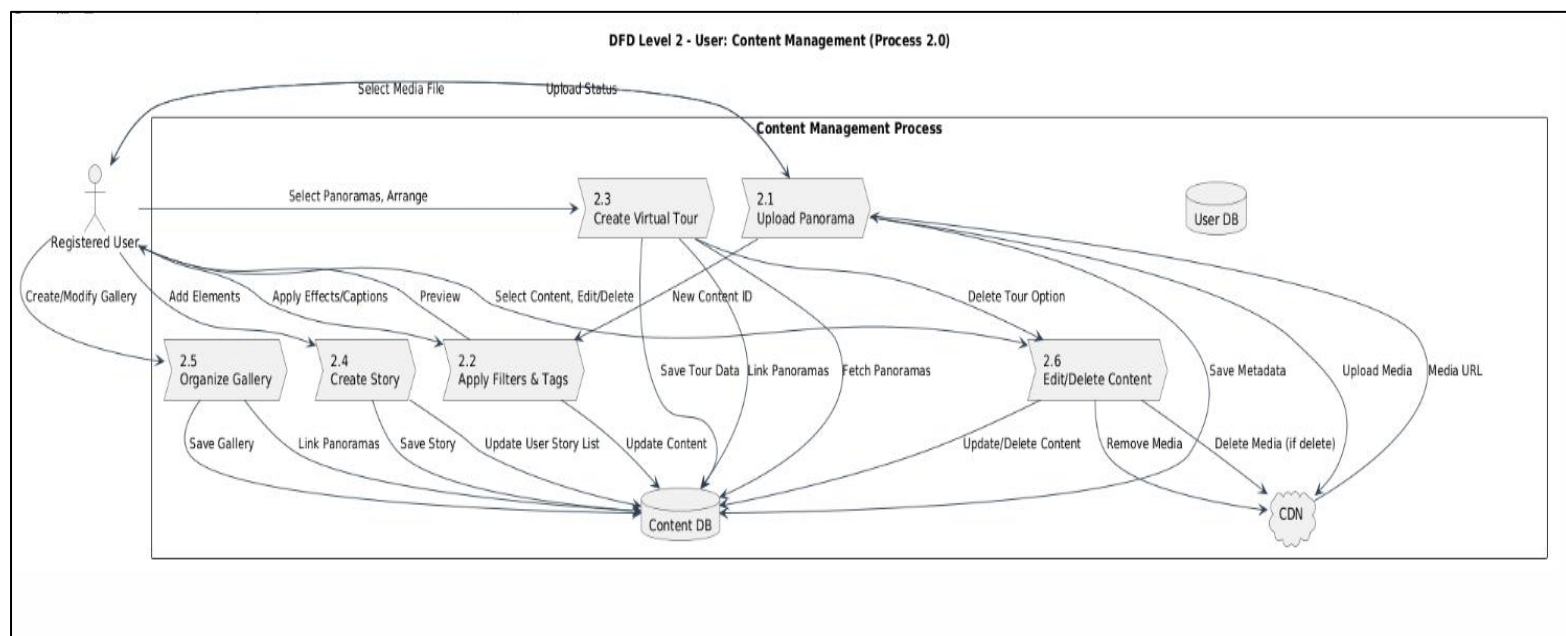
The user sends registration, content, social actions, search, analytics requests, and reports, and receives profile data, status, results, insights, and confirmation.



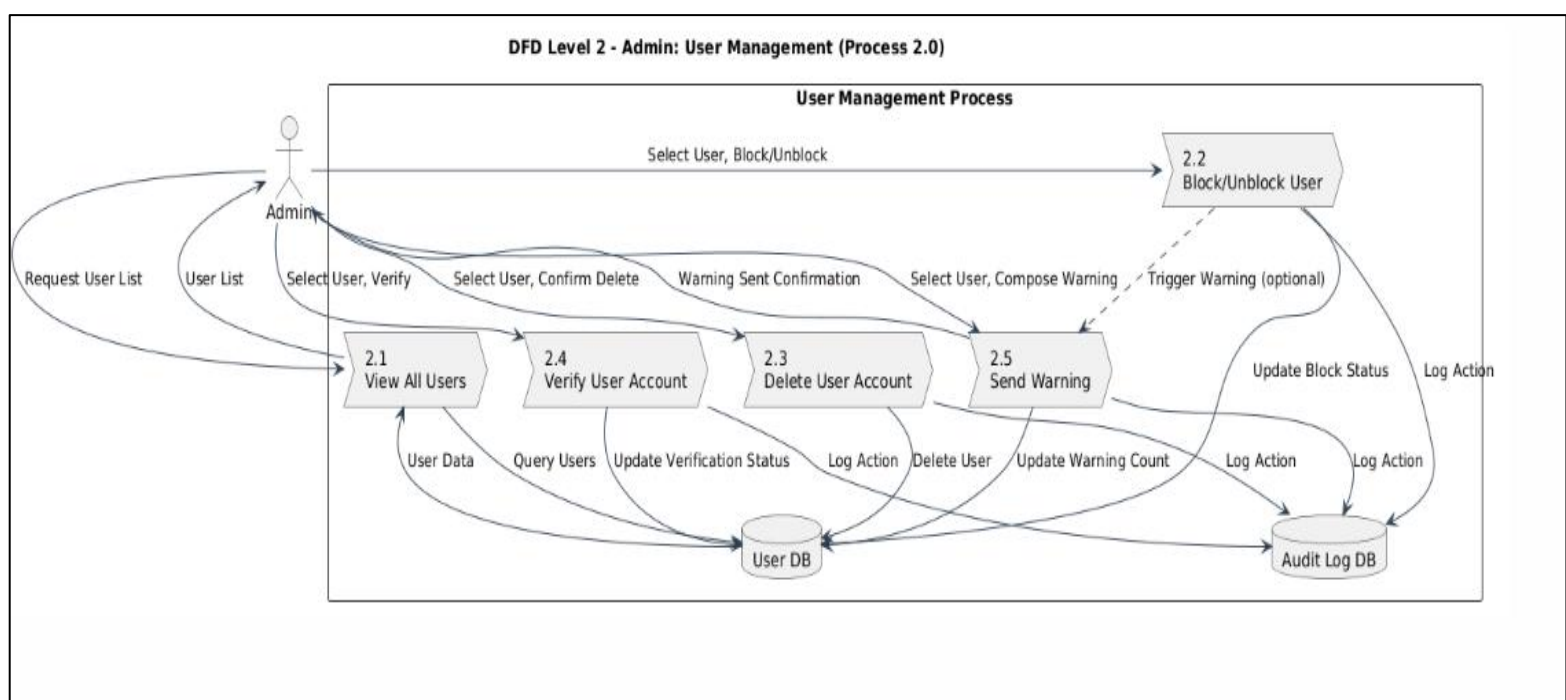
7.3 Level 1 DFD - Admin User Subsystem



7.4 Level 2 DFD – Registered User: Content Management (Detailed)



7.5 Level 2 DFD - Admin: User Management (Detailed)



8 Requirement Traceability Matrix (RTM)

Req ID	Requirement Description	Type	Priority (MoSCoW)	Design Reference	Test Case ID
FR-1	<i>User Registration: User can register with email/phone</i>	Functional	Must Have	User Management Module	TC-01
FR-2	<i>User Login & Authentication with 2FA</i>	Functional	Must Have	User Management Module	TC-02
FR-3	<i>Edit Profile & Upload Profile/Cover Image</i>	Functional	Must Have	User Management Module	TC-03
FR-4	<i>Follow/Unfollow & Block/Unblock Users</i>	Functional	Should Have	Social Module	TC-04
FR-5	<i>Upload Panorama with Filters & Tags</i>	Functional	Must Have	Content Management Module	TC-05
FR-6	<i>Create Virtual Tour by Linking Panoramas</i>	Functional	Should Have	Content Management Module	TC-06
FR-7	<i>Create Story with Text/Stickers/Filters</i>	Functional	Should Have	Content Management Module	TC-07
FR-8	<i>View Panorama with Interactive Viewer</i>	Functional	Must Have	Viewing Module	TC-08
FR-9	<i>Like, Comment, Share on Content</i>	Functional	Must Have	Social Module	TC-09
FR-10	<i>Send Direct Message with Media Attachments</i>	Functional	Should Have	Social Module	TC-10
FR-11	<i>Create Group Chat & Add/Remove Members</i>	Functional	Could Have	Social Module	TC-11

FR-12	<i>Search Users & Panoramas with Filters</i>	Functional	Should Have	Search Module	TC-12
FR-13	<i>Browse Categories & Trending Content</i>	Functional	Should Have	Search Module	TC-13
FR-14	<i>View Analytics: Engagement, Followers, Reach</i>	Functional	Should Have	Analytics Module	TC-14
FR-15	<i>Receive Notifications for Interactions</i>	Functional	Must Have	Notification Module	TC-15
FR-16	<i>Manage Privacy Settings & Post Visibility</i>	Functional	Must Have	Settings Module	TC-16
FR-17	<i>Report Inappropriate Content</i>	Functional	Must Have	Moderation Module	TC-17
FR-18	<i>Appeal Moderation Decision</i>	Functional	Could Have	Moderation Module	TC-18
FR-19	<i>Admin: View All Users & User Details</i>	Functional	Must Have	Admin Module	TC-19
FR-20	<i>Admin: Block/Unblock & Delete User Accounts</i>	Functional	Must Have	Admin Module	TC-20
FR-21	<i>Admin: Review & Remove Reported Content</i>	Functional	Must Have	Admin Module	TC-21
FR-22	<i>Admin: Approve/Reject Content</i>	Functional	Should Have	Admin Module	TC-22
FR-23	<i>Admin: View Platform Analytics & Reports</i>	Functional	Should Have	Admin Module	TC-23
FR-24	<i>Admin: Update System Configuration</i>	Functional	Could Have	Admin Module	TC-24
FR-25	<i>Admin: Broadcast Announcements & Alerts</i>	Functional	Should Have	Admin Module	TC-25
NFR-1	<i>Performance: Support 1000 concurrent users</i>	Non-Functional	Must Have	System Architecture	TC-NFR-01

NFR-2	<i>Security: Data encryption & secure authentication</i>	Non-Functional	Must Have	Security Module	TC-NFR-02
NFR-3	<i>Usability: User-friendly interface</i>	Non-Functional	Should Have	UI/UX Design	TC-NFR-03
NFR-4	<i>Compatibility: Works on Chrome, Firefox, Edge</i>	Non-Functional	Must Have	Browser Support	TC-NFR-04
NFR-5	<i>Backup & Recovery: Daily backup & restore</i>	Non-Functional	Should Have	Database Design	TC-NFR-05
NFR-6	<i>Availability: 99.9% uptime</i>	Non-Functional	Must Have	System Infrastructure	TC-NFR-06

8.1 Forward Traceability Matrix (Requirement → Design → Test)

Status	Requirement ID	Design / Module	Test Case ID
☐	FR-1 User Registration	User Management	TC-01
☐	FR-2 User Login & 2FA	User Management	TC-02
☐	FR-3 Edit Profile	User Management	TC-03
☐	FR-4 Follow/Block Users	Social Module	TC-04
☐	FR-5 Upload Panorama	Content Management	TC-05
☐	FR-6 Create Virtual Tour	Content Management	TC-06
☐	FR-7 Create Story	Content Management	TC-07
☐	FR-8 View Panorama	Viewing Module	TC-08
☐	FR-9 Like/Comment/Share	Social Module	TC-09

☐	FR-10 Direct Message	Social Module	TC-10
☐	FR-11 Group Chat	Social Module	TC-11
☐	FR-12 Search & Filters	Search Module	TC-12
☐	FR-13 Browse Trending	Search Module	TC-13
☐	FR-14 View Analytics	Analytics Module	TC-14
☐	FR-15 Notifications	Notification Module	TC-15
☐	FR-16 Privacy Settings	Settings Module	TC-16
☐	FR-17 Report Content	Moderation Module	TC-17
☐	FR-18 Appeal Decision	Moderation Module	TC-18
☐	FR-19 Admin: View Users	Admin Module	TC-19
☐	FR-20 Admin: Block/Delete Users	Admin Module	TC-20
☐	FR-21 Admin: Remove Content	Admin Module	TC-21
☐	FR-22 Admin: Approve/Reject	Admin Module	TC-22
☐	FR-23 Admin: View Analytics	Admin Module	TC-23
☐	FR-24 Admin: System Config	Admin Module	TC-24
☐	FR-25 Admin: Announcements	Admin Module	TC-25
☐	NFR-1 Performance	System Architecture	TC-NFR-01
☐	NFR-2 Security	Security Module	TC-NFR-02
☐	NFR-3 Usability	UI/UX Design	TC-NFR-03
☐	NFR-4 Compatibility	Browser Support	TC-NFR-04
☐	NFR-5 Backup & Recovery	Database Design	TC-NFR-05
☐	NFR-6 Availability	System Infrastructure	TC-NFR-06

8.2 Reverse Traceability Matrix (Test / Design → Requirement)

Status	Test Case ID	Design / Module	Mapped Requirement
☐	TC-01	User Management	FR-1
☐	TC-02	User Management	FR-2
☐	TC-03	User Management	FR-3
☐	TC-04	Social Module	FR-4
☐	TC-05	Content Management	FR-5
☐	TC-06	Content Management	FR-6
☐	TC-07	Content Management	FR-7
☐	TC-08	Viewing Module	FR-8
☐	TC-09	Social Module	FR-9
☐	TC-10	Social Module	FR-10
☐	TC-11	Social Module	FR-11
☐	TC-12	Search Module	FR-12
☐	TC-13	Search Module	FR-13
☐	TC-14	Analytics Module	FR-14
☐	TC-15	Notification Module	FR-15
☐	TC-16	Settings Module	FR-16
☐	TC-17	Moderation Module	FR-17
☐	TC-18	Moderation Module	FR-18
☐	TC-19	Admin Module	FR-19

☐	TC-20	Admin Module	FR-20
☐	TC-21	Admin Module	FR-21
☐	TC-22	Admin Module	FR-22
☐	TC-23	Admin Module	FR-23
☐	TC-24	Admin Module	FR-24
☐	TC-25	Admin Module	FR-25
☐	TC-NFR-01	System Architecture	NFR-1
☐	TC-NFR-02	Security Module	NFR-2
☐	TC-NFR-03	UI/UX Design	NFR-3
☐	TC-NFR-04	Browser Support	NFR-4
☐	TC-NFR-05	Database Design	NFR-5
☐	TC-NFR-06	System Infrastructure	NFR-6

8.3 Tracking Matrix

Req ID	Requirement	Status	Notes / Next Step
FR-1	User Registration	✓ Done	Tested and live
FR-2	User Login & 2FA	⚙ In Progress	2FA implementation pending
FR-3	Edit Profile	✓ Done	Profile image upload working
FR-4	Follow/Block Users	✓ Done	Tested
FR-5	Upload Panorama	⚙ In Progress	Filter effects pending
FR-6	Create Virtual Tour	✗ Not Started	Design phase

FR-7	Create Story	✗ Not Started	To be scheduled
FR-8	View Panorama	✓ Done	Interactive viewer working
FR-9	Like/Comment/Share	✓ Done	Tested
FR-10	Direct Message	🔄 In Progress	Media attachment pending
FR-11	Group Chat	✗ Not Started	Future release
FR-12	Search & Filters	✓ Done	Tested
FR-13	Browse Trending	🔄 In Progress	Algorithm development
FR-14	View Analytics	✗ Not Started	After core modules
FR-15	Notifications	🔄 In Progress	Push notification setup
FR-16	Privacy Settings	✓ Done	Tested
FR-17	Report Content	✓ Done	Tested
FR-18	Appeal Decision	✗ Not Started	Future release
FR-19	Admin: View Users	✓ Done	Tested
FR-20	Admin: Block/Delete Users	✓ Done	Tested
FR-21	Admin: Remove Content	✓ Done	Tested
FR-22	Admin: Approve/Reject	🔄 In Progress	Queue system pending
FR-23	Admin: View Analytics	✗ Not Started	Dashboard design phase
FR-24	Admin: System Config	✗ Not Started	After Phase 1
FR-25	Admin: Announcements	✓ Done	Broadcast working
NFR-1	Support 1000 concurrent users	✗ Not Started	Load testing pending
NFR-2	Data Encryption	✓ Done	SSL implemented
NFR-3	User-friendly UI	🔄 In Progress	UX review ongoing

<i>NFR-4</i>	<i>Browser Compatibility</i>	🔄 In Progress	Cross-browser testing
<i>NFR-5</i>	<i>Daily Backup & Recovery</i>	✓ Done	Backup scripts running
<i>NFR-6</i>	<i>99.9% Uptime</i>	🔄 In Progress	Monitoring setup